Fair market value of plan assets at 31.3.20X2	**(B)**	11,40,000
Actual return on plan assets	**(B – A)**	**2,60,000**

Question 5

Pendora Ltd. has given the following details in respect of employee benefit pension plan:

Particulars	**Amount (₹)**
The fair value of plan assets as on 01-04-20X1	5,00,000
The benefits paid out on 30-11-20X1	63,000
Inward contributions received on 30-09-20X1	1,42,000
The fair value of plan assets as on 31-03-20X2	7,50,000

On 01.04.20X1, the company made following estimates, based on its market studies and prevailing prices:

Particulars	**%**
Interest and dividend income (after tax) payable by fund	10.50
Realised gains on plan assets (after tax)	2.00
Fund administrative costs	–2.00
Expected rate of annual return (Interest is compounded annually)	10.50

You are required to find the expected and actual returns on plan assets as on 31.03.20X2 as per AS 15.

Answer

Computation of Expected and Actual Returns on Plan Assets

	₹
Return on ₹ 5,00,000 held for 12 months at 10.50%	52,500
Return on ₹ 1,42,000 for 6 months at 10.50%	7,455
Loss of interest on benefits paid for 4 months on ₹ 63,000 for 4 months @ 10.50%	(2,205)
Expected return on plan assets for 20X1-20X2	**57,750**

Fair value of plan assets as on 31st March 20X2		7,50,000
Less: Fair value of plan assets as on 1st April, 20X1	5,00,000	
Contributions received on 30.9.20X1	1,42,000	(6,42,000)
		1,08,000
Add: Benefits paid on 30th Nov 20X1		63,000
Actual return on plan assets		**1,71,000**

Question 6

As on 1st April, 20X1 the fair value of plan assets was ₹ 1,00,000 in respect of a pension plan of Zeleous Ltd. On 30th September, 20X1 the plan paid out benefits of ₹ 19,000 and received inward contributions of ₹ 49,000. On 31st March, 20X2 the fair value of plan assets was ₹ 1,50,000 and present value of the defined benefit obligation was ₹ 1,47,920. Actuarial losses on the obligations for the year 20X1-20X2 were ₹ 600.

Compounding happens half-yearly. The normal interest rate for 6 months period is 10% per annum.

On 1st April, 20X1, the company made the following estimates, based on its market studies, understanding and prevailing prices.

Particular	%
Interest & dividend income, after tax payable by the fund	9.25
Realised and unrealised gains on plan assets (after tax)	2.00
Fund administrative costs	(1.00)
Expected Rate of Return (Effective interest rate for 12 months)	10.25

You are required to find the expected and actual returns on plan assets.

Answer

Computation of Expected and Actual Returns on Plan Assets

Expected return on plan assets	₹
Return on ₹ 1,00,000 held for 12 months at 10.25%	10,250
Return on ₹ 30,000 (49,000 – 19,000) held for six months at 5% (equivalent to 10.25% annually, compounded every six months)	1,500
Expected return on plan assets for 20X1-20X2	**11,750**

Actual return on plan assets		₹
Fair value of plan assets as on 31st March, 20X2		1,50,000
Less: Fair value of plan assets as on 1st April, 20X1	1,00,000	
Contributions received	49,000	(1,49,000)
		1,000
Add: Benefits paid		19,000
Actual return on plan assets		**20,000**

Question 7

The fair value of plan assets of Anupam Ltd. was ₹ 2,00,000 in respect of employee benefit pension plan as on 1st April, 20X1. On 30th September, 20X1 the plan paid out benefits of ₹ 25,000 and received inward contributions of ₹ 55,000. On 31st March, 20X2 the fair value of plan assets was ₹ 3,00,000.

Compounding happens half-yearly. The normal interest rate for 6 months period is 10% per annum.

On 1st April, 20X1 the company made the following estimates, based on its market studies and prevailing prices.

Particular	%
Interest and dividend income (after tax) payable by fund	10.25
Realized gains on plan assets (after tax)	3.00
Fund administrative costs	(3.00)
Expected rate of return (Effective interest rate for 12 months)	10.25

Calculate the expected and actual returns on plan assets as on 31st March, 20X2, as per AS 15.

Answer

Computation of Expected Returns on Plan Assets as on 31st March, 20X2, as per AS 15

Particular	₹
Return on opening value of plan assets of ₹ 2,00,000 (held for the year) @ 10.25%	20,500
Add: Return on net gain of ₹ 30,000 (i.e. ₹ 55,000 – ₹ 25,000) during the year i.e. held for six months @ 5% (equivalent to 10.25% annually, compounded every six months)	1,500
Expected return on plan assets as on 31st March, 20X2	**22,000**

Computation of Actual Returns on Plan Assets as on 31st March, 20X2, as per AS 15

Particular		₹
Fair value of Plan Assets as on 31st March, 20X2		3,00,000
Less: Fair value of Plan Assets as on 1st April, 20X1	2,00,000	
Less: Contribution received as on 30th September, 20X1	55,000	(2,55,000)
		45,000
Add: Benefits paid as on 30th September, 20X1		25,000
Actual returns on Plan Assets as on 31st March, 20X2		**70,000**

Question 8

What is the difference between Defined Contribution Plan and Defined Benefit Plan? From the following information calculate the amount of defined benefit liability/asset:

Particulars	₹ in lakhs
Present Value of Defined Benefit Obligation as on 31-3-20X1	36.0
Fair Value of Plan asset	38.5
Past service cost not yet recognized	7.5
Present value of available future refund from the plan	6.0

Answer

(a) Difference between Defined Contribution Plan and Defined Benefit Plan:

S. No.	Defined Contribution Plan	Defined Benefit Plan
1	Fixed Contributions are paid by the employer into a separate fund and will have **no obligation to pay further contributions**.	Detailed **actuarial calculation is performed to determine the charge**.
2	The employer has no obligation to pay further contributions if the fund does not hold sufficient assets to pay all employee benefits and the **employee has to bear** the investment and actuarial **risk**.	The employer ensures that sufficient funds are available to meet the promised benefits regardless of fund performance and the actuarial and investment **risk fall on the employer**.

(b) Computation of defined benefit liability /Asset:

Particulars	₹ in lakhs
Present value of the defined benefit obligation as on 31.3.20X1	36.00

Less: Past service cost not yet recognized	(7.50)
	28.50
Less: The fair value of plan assets	(38.50)
Defined benefit Asset	**10.00**

In case where fair value of plan assets is high, it may so happen that the net amount under defined benefit liability turns negative (giving rise to net assets).

As per AS 15 the enterprise, in such a situation, should measure the resulting asset at the **lower of**:

the amount so determined, i.e. **₹ 10 lakh**; and the present value of available future refunds from the plan i.e. **₹ 6 lakh**. Therefore, defined benefit asset will be **recognised at ₹ 6 lakhs.**

Question 9

Rock Star Ltd. discontinues a business segment. Under the agreement with employee's union, the employees of the discontinued segment will earn no further benefit. This is a curtailment without settlement, because employees will continue to receive benefits for services rendered before discontinuance of the business segment. Curtailment reduces gross obligation for various reasons including change in actuarial assumptions made before curtailment. If the benefits are determined based on the last pay drawn by employees, the gross obligation reduces after the curtailment because the last pay earlier assumed is no longer valid.

Rock Star Ltd. estimates the share of unamortized service cost that relates to the part of the obligation at ₹ 18 (10% of ₹ 180). Calculate the gain from curtailment and liability after curtailment to be recognised in the balance sheet of Rock Star Ltd. on the basis of given information:

(a) Immediately before the curtailment, gross obligation is estimated at ₹ 6,000 based on current actuarial assumption.

(b) The fair value of plan assets on the date is estimated at ₹ 5,100.

(c) The unamortized past service cost is ₹ 180.

(d) Curtailment reduces the obligation by ₹ 600, which is 10% of the gross obligation.

Answer

Gain from curtailment is estimated as under:

Particular	**₹**
Reduction in gross obligation	600
Less: Proportion of unamortised past service cost	(18)
Gain from curtailment	**582**

The liability to be recognised after curtailment in the balance sheet is estimated as under:

Particular	**₹**
Reduced gross obligation (90% of ₹ 6,000)	5,400
Less: Fair value of plan assets	(5,100)
	300
Less: Unamortised past service cost (90% of ₹ 180)	(162)
Liability to be recognised in the balance sheet	**138**

CHAPTER - 16
AS 16: Borrowing Costs

Treatment of Borrowing Costs

Question 1

Take Ltd. has borrowed ₹ 30 lakhs from State Bank of India during the financial year 20X1-20X2. The borrowings are used to invest in shares of Give Ltd., a subsidiary company of Take Ltd., which is implementing a new project, estimated to cost ₹ 50 lakhs. As on 31st March, 20X2, since the said project was not complete, the directors of Take Ltd. resolved to capitalise the interest accruing on borrowings amounting to ₹ 4 lakhs and add it to the cost of investments. Comment.

Answer

As per **AS 13** (Revised) "Accounting for Investments", the cost of investment includes acquisition charges such as brokerage, fees and duties. In the present case, Take Ltd. has used borrowed funds for purchasing shares of its subsidiary company Give Ltd. ₹ 4 lakhs **interest** payable by Take Ltd. to State Bank of India cannot be called as acquisition charges, therefore, **cannot be constituted as cost of investment**.

Further, as per para 3 of **AS 16** "Borrowing Costs", a qualifying asset is an asset that necessarily takes a substantial period of time to get ready for its intended use or sale. Since, shares are ready for its intended use at the time of sale, it **cannot be considered as qualifying asset** that can enable a company to add the borrowing cost to investments. Therefore, the directors of Take Ltd. **cannot capitalise** the **borrowing cost** as part of cost of investment. Rather, it has to be **charged to the Statement of Profit and Loss** for the year ended 31st March, 20X2.

Period of Capitalisation of Borrowing Cost on Qualifying Asset

Question 2

H Ltd. incurs borrowing costs for the purpose of construction of a qualifying asset for its own use. The construction gets completed on May 31, 20X1. However, decoration work is under process which is expected to be completed by November 20X1 after which H Ltd. will be able to start using the said asset for its own use. H Ltd. wants to capitalize the eligible borrowing costs incurred up to November 20X1.

Answer

The capitalization of borrowing costs shall cease when substantially all the activities necessary to prepare the qualifying assets for its intended use or sale is completed.

In the given case, H Ltd. should **capitalize borrowing costs only up to May 31, 20X1**. The **borrowing cost incurred thereafter cannot be capitalized** as the asset was ready for its intended use on May 31, 20X1. The fact that decoration work was being carried out should not be considered as the asset was ready for its intended use on May 31, 20X1.

Question 3

ABC Ltd. is in the process of getting an entertainment park constructed. For this purpose, it has taken loan from a bank. The said park consists of several rides and facilities, each of which can be used individually. Three

fourth part of the park has been constructed and can be opened up for public, while construction on the remaining part is continuing. Whether the capitalization of borrowing cost should continue for the whole park until construction continues?

Answer

ABC Ltd. is in process of constructing an entertainment park which consists of several rides and facilities that can operate independently for their intended use. Even though the park as whole is not complete, the individual facilities are ready for their intended use.

The cessation of capitalization depends upon the nature of the qualifying assets, particularly where the qualifying assets consists of various parts. There are qualifying assets where each part is capable of being used while the construction continues on other parts. There are qualifying assets where all parts have to be completed before any earlier completed part can be put to use.

Since in the given scenario, the individual facilities are capable of operating independently and are ready for their intended use, therefore the **borrowing costs shall cease to be capitalized for the three-fourth part** of the project.

If Multiple Assets are Purchased/Constructed using Specific Borrowing

Question 4

PRM Ltd. obtained a loan from a bank for ₹ 120 lakhs on 30.04.20X1. It was utilised as follows:

Particulars	₹ in lakhs
Construction of a shed	50
Purchase of a machinery	40
Working Capital	20
Advance for purchase of truck	10

Construction of shed was completed in March 20X2. The machinery was installed on the date of acquisition. Delivery of truck was not received. Total interest charged by the bank for the year ending 31.03.20X2 was ₹ 18 lakhs. Show the treatment of interest.

Answer

According to **AS 16 '**Borrowing costs', qualifying asset is an asset that necessarily takes substantial period of time to get ready for its intended use.

Borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset should be capitalised as part of the cost of that asset. Other borrowing costs should be recognised as an expense in the period in which they are incurred.

The treatment of interest can be shown as: **₹ in lakhs**

Purpose	Nature	Interest to be capitalized	Interest to be charged to profit and loss account
Construction of a shed	Qualifying asset	(50/120) x 18 = 7.5	
Purchase of a machinery	Not a qualifying asset		(40/120) x 18 = 6
Working Capital	Not a qualifying asset		(20/120) x 18 = 3

Advance for purchase of truck	Not a qualifying asset		(10/120) x 18 = 1.5
Total		**7.5**	**10.5**

Question 5

The company has obtained Institutional Term Loan of ₹ 580 lakhs for modernisation and renovation of its Plant & Machinery. Plant & Machinery acquired under the modernisation scheme and installation completed on 31st March, 20X2 amounted to ₹ 406 lakhs, ₹ 58 lakhs has been advanced to suppliers for additional assets (additional asset has taken substantial period of time in its installation) and the balance loan of ₹ 116 lakhs has been utilised for working capital purpose. The Accountant is on a dilemma as to how to account for the total interest of ₹ 52.20 lakhs incurred during 20X1-20X2 on the entire Institutional Term Loan of ₹ 580 lakhs.

Answer

As per para 6 of AS 16 'Borrowing Costs', borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset should be capitalised as part of the cost of that asset. Other borrowing costs should be recognised as an expense in the period in which they are incurred.

A qualifying asset is an asset that necessary takes a substantial period of time to get ready for its intended use or sale.

The treatment for total interest amount of ₹ 52.20 lakhs can be given as: **₹ in lakhs**

Purpose	Nature	Interest to be capitalised	Interest to be charged to profit and loss account
Modernisation and renovation of plant and machinery	Qualifying asset	(406/580) x 52.2 = 36.54	
Advance to supplies for additional assets	Qualifying asset	(58/580) x 52.2 = 5.22	
Working Capital	Non qualifying asset		(116/580) x 52.2 = 10.44
Total		**41.76**	**10.44**

Question 6

On 1st April, 20X1, Amazing Construction Ltd. obtained a loan of ₹ 32 crores to be utilised as under:

(i)	Construction of sealink across two cities (work was held up totally for a month during the year due to high water levels)	₹ 25 crores
(ii)	Purchase of equipments and machineries	₹ 3 crores
(iii)	Working capital	₹ 2 crores
(iv)	Purchase of vehicles	₹ 50,00,000
(v)	Advance for tools/cranes etc.	₹ 50,00,000
(vi)	Purchase of Technical know-how	₹ 1 crores
(vii)	Total Interest charged by the bank for the year ending 31st March 20X2	₹ 80,00,000

Show the treatment of interest by Amazing Construction Ltd.

Answer

According to AS 16 'Borrowing costs', qualifying asset is an asset that necessarily takes substantial period of time to get ready for its intended use.

Borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset should be capitalised as part of the cost of that asset. Other borrowing costs should be recognised as an expense in the period in which they are incurred.

The treatment of interest by Amazing Construction Ltd. can be shown as:

Purpose	Qualifying Asset	Interest to be capitalized	Interest to be charged to P&L A/c	Calculation
Construction of sea-link	Yes	62,50,000		[80,00,000 x (25/32)]
Purchase of equipment and machineries	No		7,50,000	[80,00,000 x (3/32)]
Working capital	No		5,00,000	[80,00,000 x (2/32)]
Purchase of vehicles	No		1,25,000	[80,00,000 x (0.5/32)]
Advance for tools, cranes etc.	No		1,25,000	[80,00,000 x (0.5/32)]
Purchase of technical know-how	No		2,50,000	[80,00,000 x (1/32)]
Total		**62,50,000**	**17,50,000**	

Note: It is assumed that work held up for a month due to high water level is normal during the construction of sealink and capitalization of borrowing cost should not be suspended for necessary temporary delay.

Question 7

Expert Limited issued 12% secured debentures of ₹ 100 lakhs on 01.06.20X1. Money raised from debentures to be utilized as under:

Intended Purpose	Amount (₹ in lakhs)
Construction of factory building	40
Working Capital	30
Purchase of Machinery	15
Purchase of Furniture	2
Purchase of truck	13

Additional Information:

(i) Interest on debentures for the Financial Year 20X1-X2 was paid by the Company.

(ii) During the year, the company invested idle fund of ₹ 5 lakhs (out of the money raised from debentures) in Bank's fixed deposit and earned interest of ₹ 50,000.

(iii) In March, 20X2 construction of factory building was not completed (it is expected that it will take another 6 months).

(iv) In March 20X2, Machinery was installed and ready for its intended use.

(v) Furniture was put to use at the end of March 20X2.

(vi) Truck is going to be received in April, 20X2.

You are required to show the treatment of interest as per AS 16 in respect of borrowing cost for the year ended 31st March, 20X2 in the Books of Expert Limited.

Answer

According to AS 16 "Borrowing Costs", a qualifying asset is an asset that necessarily takes a substantial period of time to get ready for its intended use. As per the Standard, borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset should be capitalized as part of the cost of that asset. The amount of borrowing costs eligible for capitalization should be determined in accordance with this Standard. Other borrowing costs should be recognized as an expense in the period in which they are incurred. It also states that to the extent that funds are borrowed specifically for the purpose of obtaining a qualifying asset, the amount of borrowing costs eligible for capitalization on that asset should be determined as the actual borrowing costs incurred on that borrowing during the period less any income on the temporary investment of those borrowings.

Thus, eligible borrowing cost = ₹ 10,00,000 (100 lakhs x 12% x 10/12) – ₹ 50,000 = ₹ 9,50,000

Particulars	**Nature of assets**	**Interest to be capitalized (₹)**	**Interest to be charged to Profit & Loss Account (₹)**
Construction of factory building	Qualifying Asset	9,50,000 x 40/100 = ₹ 3,80,000	NIL
Purchase of Machinery	Not a Qualifying Asset	NIL	9,50,000 x 15/100 = 1,42,500
Purchase of furniture	Not a Qualifying Asset	NIL	9,50,000 x 2/100 = 19,000
Purchase of truck	Not a Qualifying Asset	NIL	9,50,000 x 13/100 = 1,23,500
Working Capital	Not a Qualifying Asset	NIL	9,50,000 x 30/100 = 2,85,000
Total		**₹ 3,80,000**	**₹ 5,70,000**

Question 8

Harish Construction Company is constructing a huge building project consisting of four phases. It is expected that the full building will be constructed over several years but Phase I and Phase II of the building will be started as soon as they are completed.

Following is the detail of the work done on different phases of the building during the current year:

(₹ in lakhs)

Particulars	**Phase I**	**Phase II**	**Phase III**	**Phase IV**
Cash expenditure	10	30	25	30
Building purchased	24	34	30	38
Total expenditure	34	64	55	68
Total expenditure of all phases				221
Loan taken @ 15% at the beginning of the year				200

During mid of the current year, Phase I and Phase II have become operational. Find out the total amount to be capitalized and to be expensed during the year.

Answer

Computation of amount to be capitalized

No.	**Particulars**	**₹**
1.	Interest expense on loan ₹ 2,00,00,000 at 15%	30,00,000

2.	Total cost of Phases I and II (₹ 34,00,000 + ₹ 64,00,000)	98,00,000
3.	Total cost of Phases III and IV (₹ 55,00,000 + ₹ 68,00,000)	1,23,00,000
4.	Total cost of all 4 phases	2,21,00,000
5.	Total loan	2,00,00,000
6.	Interest on loan used for Phases I & II, based on proportionate Loan amount = (30,00,000/2,21,00,000) x 98,00,000	3,30,317 (approx.)
7.	Interest on loan used for Phases III & IV, based on proportionate Loan amount = (30,00,000/2,21,00,000) x 1,23,00,000	16,69,683 (approx.)

Accounting treatment

For Phase I and Phase II

Since Phase I and Phase II have become operational at the mid of the year, **half of the interest amount of ₹ 6,65,158.50** (i.e. ₹ 13,30,317/2) relating to Phase I and Phase II should be **capitalized** (in the ratio of asset costs 34:64) and added to respective assets in Phase I and Phase II and remaining half of the interest amount of ₹ 6,65,158.50 (i.e. ₹ 13,30,317/2) relating to Phase I and Phase II should be expensed during the year.

For Phase III and Phase IV

Interest of ₹ 16,69,683 relating to Phase III and Phase IV should be held in Capital Work-in-Progress till assets construction work is completed, and thereafter **capitalized** in the ratio of cost of assets. No part of this interest amount should be charged/expensed off during the year since the work on these phases has not been completed yet.

If Single Qualifying Asset is Purchased/Constructed using Specific & General Borrowing

Question 9

Rainbow Limited borrowed an amount of ₹ 150 crores on 1.4.20X1 for construction of boiler plant @ 11% p.a. The plant is expected to be completed in 4 years. Since the weighted average cost of capital is 13% p.a., the accountant of Rainbow Ltd. capitalized ₹ 19.50 crores for the accounting period ending on 31.3.20X2. Due to surplus fund out of ₹ 150 crores, income of ₹ 3.50 crores was earned and credited to profit and loss account. Comment on the above treatment of accountant with reference to relevant accounting standard.

Answer

Para 10 of AS 16 'Borrowing Costs' states "To the extent that funds are borrowed specifically for the purpose of obtaining a qualifying asset, the amount of borrowing costs eligible for capitalization on that asset should be determined as the actual borrowing costs incurred on that borrowing during the period less any income on the temporary investment of those borrowings."

The capitalization rate should be the weighted average of the borrowing costs applicable to the borrowings of the enterprise that are outstanding during the period, other than borrowings made specifically for the purpose of obtaining a qualifying asset.

Thus, the treatment of accountant of Rainbow Ltd. is incorrect.

Amount of borrowing costs capitalized should be calculated as follows:

Particulars	₹ in crores
Actual interest for 20X1-20X2 (11% of ₹ 150 crores)	16.50

Less: Income on temporary investment from specific borrowings	(3.50)
Borrowing costs to be capitalized during year 20X1-20X2	**13.00**

Question 10

ABC Limited has started construction of an asset on 1st December, 20X1, which continues till 31st March, 20X2 (and is expected to go beyond a year). The entity has not taken any specific borrowings to finance the construction of the asset but has incurred finance costs on its general borrowings during the construction period. The directly attributable expenditure at the beginning of the month on this asset was ₹ 10 lakh in December 20X1 and ₹ 4 lakh in each of the months of January to March 20X2. At the beginning of the year, the entity had taken Inter Corporate Deposits of ₹ 20 lakh at 9% rate of interest and had an overdraft of ₹ 4 lakh, which increased to ₹ 8 lakh on 1st March, 20X2. Interest was paid on the overdraft at 10% until 1st January, 20X2 and then the rate was increased to 12%. You are required to calculate the annual capitalization rate for computation of borrowing cost in accordance with AS 16 'Borrowing Costs'.

Answer

Calculation of capitalization rate on borrowings other than specific borrowing

Nature of general borrowings	Period of outstanding balance	Amount of loan (₹)	Rate of interest p.a.	Weighted average amount of interest (₹)
	a	b	c	d = [(b x c) x (a/12)]
9% Debentures	12 months	20,00,000	9%	1,80,000
Bank Overdraft	9 months	4,00,000	10%	30,000
	2 months	4,00,000	12%	8,000
	1 months	8,00,000	12%	8,000
		36,00,000		**2,26,000**

Weighted average cost of borrowings = {20,00,000 x (12/12)} + {4,00,000 x (11/12)} + {8,00,000 x (1/12)}
= 24,33,334

Capitalisation rate = [(Weighted average amount of interest/Weighted average of general borrowing) x 100]
= [(2,26,000/24,33,334) x 100] = 9.29% p.a.

Question 11

X Ltd. began construction of a new building on 1st January, 20X1. It obtained ₹ 1 lakh special loan to finance the construction of the building on 1st January, 20X1 at an interest rate of 10%. The company's other outstanding two non-specific loans were:

Amount	Rate of Interest
₹ 5,00,000	11%
₹ 9,00,000	13%

The expenditures that were made on the building project were as follows:

	₹
1st January 20X1	2,00,000

1st April 20X1	2,50,000
1st July 20X1	4,50,000
1st December 20X1	1,20,000

Building was completed by 31st December 20X1. Following the principles prescribed in AS 16 'Borrowing Cost,' calculate the amount of interest to be capitalised and pass one Journal Entry for capitalising the cost and borrowing cost in respect of the building.

Answer

(i) Calculation of weighted average interest rate other than for specific borrowings

Amount of loan (₹)	Rate of interest	Amount Interest (₹)
5,00,000	11%	55,000
9,00,000	13%	1,17,000
14,00,000		1,72,000

Weighted average rate of interest = [(1,72,000/14,00,000) x 100] = **12.285% (approx.)**

(ii) Computation of weighted average accumulated expenses

	₹
₹ 2,00,000 x 12/12	2,00,000
₹ 2,50,000 x 9/12	1,87,500
₹ 4,50,000 x 6/12	2,25,000
₹ 1,20,000 x 1/12	10,000
	6,22,500

(iii) Interest on weighted average accumulated expenses

	₹
Specific borrowings (₹ 1,00,000 x 10%)	10,000
Non-specific borrowings [(₹ 6,22,500 – ₹ 1,00,000) x 12.285%]	64,189
Amount of interest to be capitalized	**74,189**

(iv) Total expenses to be capitalized for building

	₹
Cost of building ₹ (2,00,000 + 2,50,000 + 4,50,000 + 1,20,000)	10,20,000
Add: Amount of interest to be capitalized	74,189
Amount of interest to be capitalized	**10,94,189**

(v) Journal Entry

Date		Debit (₹)	Credit (₹)
31.12.20X1	Building account Dr.	10,94,189	
	To Bank account		10,94,189
	(Being amount of cost of building and borrowing cost there on capitalised)		

Question 12

On 1st April, 20X1, Green Limited started the construction of an Office Building (qualified asset). The land under the building is regarded as a separate asset and is not a part of qualifying asset.

For the purpose of construction of building, the company raised a specific loan of ₹ 14 lakhs from a Bank at an interest rate of 12% per annum. An interest income of ₹ 15,000 was earned on this loan while it was held in anticipation of payments.

The company's other outstanding loans on 1st April, 20X1 were as follows:

Amount of Loan	Rate of Interest per annum
₹ 20,00,000	15%
₹ 30,00,000	8%

The construction of building started on 1st April, 20X1 and was completed on 31st January, 20X2 when it was ready for its intended use. Up to the date of completion of the building, the following payments were made to the contractor:

Payment date	Amount in ₹
1st April, 20X1	4,00,000
1st August, 20X1	10,00,000
1st December, 20X1	25,00,000
31st January, 20X2	5,00,000

The life of building is estimated to be 20 years and depreciation is calculated on straight line method.

You are required to:

(i) Calculate the amount of borrowing cost to be capitalized.

(ii) Pass initial journal entry to recognise the cost of building.

(iii) Depreciation on building for the year ending 31st March, 20X2.

(iv) Carrying value of building as on 31st March, 20X2.

Answer

(i) Computation of borrowing cost to be capitalized for specific borrowings and general borrowings based on weighted average accumulated expenses

Date of expenditure	Amount spent	Financed through	Calculation	₹
1.4.20X1	4,00,000	Specific borrowing	4,00,000 x 12% x 10/12	40,000
1.8.20X1	10,00,000	Specific borrowing	10,00,000 x 12% x 10/12	1,00,000
1.12.20X1	25,00,000	General borrowing	25,00,000 x 10.8% x 2/12	45,000
31.1.20X2	5,00,000	General borrowing	5,00,000 x 10.8% x 0/12	Nil
				1,85,000
Less: interest income on borrowing				(15,000)
Total amount borrowing cost to be capitalized				**1,70,000**

(ii) Journal Entry

Date			Debit (₹)	Credit (₹)
31.1.20X2	Building account	Dr.	45,70,000	

	To Bank account (Being amount of cost of building and borrowing cost there on capitalised)		45,70,000

(iii) Depreciation on building for the year ending 31.3.20X2

Cost of building = 45,70,000

Life of building = 20 years

Depreciation = (45,70,000/20) x 2/12 = 38,083.33

(iv) Carrying Value of Building on 31st March 20X2:

Carrying Value = Cost of Building – Accumulated Depreciation

= 45,70,000 – 38,083.33

= 45,31,917

Working Notes:

1. Calculation of capitalization rate on borrowings other than specific borrowings

Amount of loan (₹)	Rate of interest	Amount of interest (₹)
20,00,000	15%	3,00,000
30,00,000	8%	2,40,000
50,00,000		**5,40,000**
Weighted average rate of interest = (5,40,000/50,00,000 x 100)		10.8%

2. Total expenses to be capitalized for building

	₹
Cost of building ₹ (4,00,000 + 10,00,000 + 25,00,000 + 5,00,000)	44,00,000
Add: Amount of interest to be capitalized	1,70,000
	45,70,000

Exchange Difference on Foreign Currency Borrowing to be Treated as Borrowing Cost

Question 13

ABC Builders Limited had borrowed a sum of US $ 15,00,000 at the beginning of Financial year 20X1-X2 for its residential project at London Interbank Offered Rate (LIBOR) + 4 %. The interest is payable at the end of the Financial Year. At the time of availing the loan, the exchange rate was ₹ 72 per US $ and the rate as on 31st March, 20X2 was ₹ 76 per US $. If ABC Builders Limited borrowed the loan in Indian Rupee equivalent, the pricing of loan would have been 9.50%. Compute Borrowing Cost and exchange difference for the year ending 31st March, 20X2 as per applicable Accounting Standards. (Applicable LIBOR is 1%).

Answer

(i) Interest for the period 20X1-X2 = US $ 15 lakhs x 5% x ₹ 76 per US $ = ₹ 57 lakhs

(ii) Increase in the liability towards the principal amount = US $ 15 lakhs x ₹ (76 – 72) = ₹ 60 lakhs

(iii) Interest that would have resulted if the loan was taken in Indian currency = US $ 15 lakhs x ₹ 72 x 9.5% = ₹ 102.60 lakhs

(iv) Difference between interest on local currency borrowing and foreign currency borrowing = ₹ 102.60 lakhs less ₹ 57 lakhs = ₹ 45.60 lakhs.

Therefore, **out of ₹ 60 lakhs** increase in the liability towards principal amount, only **₹ 45.60 lakhs will be considered as the borrowing cost**. Thus, **total borrowing cost would be ₹ 102.60 lakhs** being the aggregate of interest of ₹ 57 lakhs on foreign currency borrowings plus the exchange difference to the extent of difference between interest on local currency borrowing and interest on foreign currency borrowing of ₹ 45.60 lakhs.

Hence, ₹ 102.60 lakhs would be considered as the borrowing cost to be accounted for as per AS 16 "Borrowing Costs" and the **remaining ₹ 14.4 lakhs (60 – 45.60) would be considered as the exchange difference** to be accounted for as per AS 11 "The Effects of Changes in Foreign Exchange Rates".

Miscellaneous

Question 14

Glen Ltd. began construction of a new building on 1st January, 20X1. On 1st April, 20X1, following two loans were obtained to fund the construction cost:

(i) Loan of ₹ 60,00,000 from Data Bank Ltd. was taken at interest rate of 8% per annum. This loan was fully utilized for construction of the new building.

(ii) Loan of ₹ 20,00,000 from Satya Bank Ltd. Out of this, loan amount of ₹ 6,00,000 was utilized for working capital purpose. Total interest of ₹ 1,92,000 were paid to Satya Bank Ltd. for the financial year 20X1-X2.

Construction of the new building was completed on 31st January, 20X2 and was ready for its intended use on the same date.

None of the loan was repaid during the year. The building is a qualifying asset for the purpose of AS-16.

Out of loan from Data Bank Ltd., surplus funds were temporarily invested for the short period of time. This temporary investment earned interest of ₹ 30,000.

You are required to calculate the amount of interest to be capitalized and to be charged to profit and loss account from the total interest incurred as borrowing cost during the year 20X1-X2 (as per AS-16).

Answer

According to AS 16 "Borrowing Costs", borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset should be capitalized as part of the cost of that asset. The amount of borrowing costs eligible for capitalization should be determined in accordance with this Standard. Other borrowing costs should be recognised as an expense in the period in which they are incurred.

The standard also states that to the extent that funds are borrowed specifically for the purpose of obtaining a qualifying asset, the amount of borrowing costs eligible for capitalization on that asset should be determined as the actual borrowing costs incurred on that borrowing during the period less any income on the temporary investment of those borrowings.

Thus, eligible borrowing cost on Loan of data bank to be capitalized = ₹ (60,00,000 x 8%) x 10/12 – ₹ 30,000 = ₹ 4,00,000 – ₹ 30,000 = ₹ 3,70,000

Loan	Particulars	Nature of assets	Interest to be Capitalized (₹)	Interest to be charged to Profit & Loss Account (₹)
Data bank	Construction of factory building	Qualifying Asset	₹ 3,70,000	(4,80,000 – 4,00,000) = ₹ 80,000

Satya Bank	Construction of factory building	Qualifying Asset	(1,92,000 x 14/20) x 10/12 = ₹ 1,12,000	(1,92,000 x 14/20) x 2/12 = ₹ 22,400
Satya Bank	Working Capital	Not a Qualifying Asset	NIL	(1,92,000 x 6/20) = ₹ 57,600
Total			**₹ 4,82,000**	**₹ 1,60,000**

Note: Loan from Satya bank is considered to be specific borrowings.

Question 15

On 1st April, 20X1 Workhouse Limited took a loan from a Financial Institution for ₹ 25,00,000 for the construction of Building. The rate of interest is 12%.

In addition to above loan, the company has taken multiple borrowings as follows:

(i) 8% Debentures ₹ 15,00,000

(ii) 15% Term Loan ₹ 30,00,000

(iii) 10% Other Loans ₹ 18,00,000

The company has utilised the above funds in construction/purchase of the following assets:

(i) Building ₹ 70,00,000

(ii) Furniture ₹ 22,00,000

(iii) Plant & Machinery ₹ 90,00,000

(iv) Factory Shed ₹ 43,00,000

The construction of Building, Plant & Machinery and Factory Shed was completed on 31st March 20X2. Readymade Furniture was purchased directly from the market. The factory was ready for production on 1st April 20X2.

You are required to calculate the borrowing cost for both qualifying and non-qualifying assets.

Answer

Interest to be Capitalized (on qualifying asset)

			₹
i.	On financial institution Borrowings	25,00,000 x 12%	3,00,000
ii.	On other borrowings	(W.N.1)	6,67,500
iii.	Amount of interest to be Capitalised	(i + ii)	9,67,500

Interest transferred to P&L (on non-qualifying asset)

			₹
i.	On other Borrowings	(W.N.1)	82,500

Working note:

1. Treatment of interest under AS 16 on other borrowings

S.N.	Particulars	Qualifying asset	Expenses Incurred	Share in borrowings	Interest-capitalized	Interest-charged to P&L A/c
i.	Building	Yes	45,00,000	7,50,000 x 45/200	1,68,750	–
ii.	Furniture	No	22,00,000	7,50,000 x 22/200	–	82,500

iii.	Plant & Machinery	Yes	90,00,000	7,50,000 x 90/200	3,37,500	–
iv.	Factory shed	Yes	43,00,000	7,50,000 x 43/200	1,61,250	–
Total			**2,00,00,000**		**6,67,500**	**82,500**

2. Total amount of Interest on other borrowings

Particulars	Amount of loan (a)	Rate of interest (b)	Amount of interest (c) = (a) x (b)
Debentures	15,00,000	8%	1,20,000
Term loan	30,00,000	15%	4,50,000
Other loans	18,00,000	10%	1,80,000
	63,00,000		**7,50,000**

Question 16

How will interest be capitalized when qualifying assets are funded by borrowings in the nature of bonds that are issued at a discount? X Ltd. issued in year 1, a 3 year 10% p.a. (interest paid annually) bond with a face value of ₹ 1,00,000 at a price of ₹ 90,000 to finance a qualifying asset which is ready for intended use at the end of year 2. Compute the amount of borrowings costs to be capitalized if the company uses for amortization of discount straight line basis.

Answer

As per **AS 16**, "Borrowing costs are **interest and other costs incurred** by an enterprise in connection with the borrowing of funds". Further, as per para 4 (b) of the standard, "amortization of discounts or premiums relating to borrowings" as a component of borrowing costs. Thus, the borrowing costs comprise the periodic interest payable on the bonds in question and the amount of discount amortised during the period.

Paragraph 6 of the Statement, inter-alia, states that "Borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset should be capitalized as part of the cost of that asset".

Further, paragraph 19 states that "Capitalisation of borrowing costs should cease when substantially all the activities necessary to prepare the qualifying asset for its intended use or sale are complete". Thus, only that portion of the amortised discount should be capitalised as part of the cost of a qualifying asset which relates to the period during which acquisition, construction or production of the asset takes place.

Straight line basis

Year	Interest	Amortisation of discount	Total borrowing costs
Year 1	10,000	3,333	13,333
Year 2	10,000	3,333	13,333
Year 3	10,000	3,334	13,334

In the above case, the amount of borrowing costs capitalized would be ₹ 13,333 in Year 1 and Year 2. The borrowing costs of ₹ 13,334 incurred in Year 3 would be expensed since the asset is ready for its intended use at the end of Year 2.

Question 17

Loyal Ltd. has undertaken a project for expansion of capacity as per the following details:

	Plan (₹)	Actual (₹)
October, 20X1	5,00,000	4,00,000
November, 20X1	6,50,000	7,95,000
December, 20X1	20,00,000	–
January, 20X2	2,00,000	50,000
February, 20X2	9,00,000	2,00,000
March, 20X2	10,00,000	12,00,000

The company pays to its bank interest at a rate of 15% p.a., which is debited on a monthly basis. During the half year, company had ₹ 20 lakh overdraft up to 31st December, surplus cash in January and again overdraft of ₹ 14 lakh from 1.2.20X2 and ₹ 30 lakh from 1.3.20X2. The company had a strike during December and hence could not continue the work during said period. However, the substantial administrative work related to the project was continued. Onsite work was again commenced on 1st January and all the work were completed on 31st March. Assume that expenditure was incurred on 1st day of each month.

Calculate interest to be capitalized giving reason wherever necessary. Assume overdraft will be less, if there is no capital expenditure.

Answer

Month	Actual Expenditure (₹)	Interest on outstanding amount @ 15% p.a.	Interest capitalized (₹)	Outstanding Cumulative amount (₹)
October, 20X1	4,00,000	4,00,000 x 15% x 1/12	5,000	4,00,000 + 5,000 = 4,05,000
November, 20X1	7,95,000	(4,05,000 + 7,95,000) x 15% x 1/12	15,000	4,05,000 + 7,95,000 + 15,000 = 12,15,000
December, 20X1	–	(12,15,000) x 15% x 1/12	15,188	12,15,000 + 15,188 = 12,30,188
January, 20X2	50,000	–	–	12,30,188 + 50,000 = 12,80,188
February, 20X2	2,00,000	14,00,000 x 15% x 1/12	17,500	12,80,188 + 2,00,000 + 17,500 = 14,97,688
March, 20X2	12,00,000	(14,97,688 + 12,00,000) x 15% x 1/12	33,721	14,97,688 + 12,00,000 + 33,721 = 27,31,409
	26,45,000		**86,409**	

Note:

1. As per para 18 of AS 16, 'Borrowing Cost', capitalisation of borrowing costs is not normally suspended during a period when substantial technical and administrative work is being carried out. Therefore, the interest for that period i.e. for the month of December has also been capitalized.
2. During January, the company did not incur any interest as there was surplus cash in January. Therefore, no amount should be capitalized during January as per para 14(b) of AS 16.
3. During February, actual overdraft (borrowings) was ₹ 14 lakh only. Hence, interest of ₹ 17,500 on ₹ 14,00,000 has been calculated even though actual expenditure on project exceeds ₹ 14 lakh.

CHAPTER - 17
AS 17: Segment Reporting

Reportable Segments

Question 1

XYZ Ltd. has 5 business segments. Profit/Loss of each of the segments for the year ended 31st March, 20X2 have been provided below. You are required to identify from the following whether reportable segments or not reportable segments, on the basis of "profitability test" as per AS-17.

Segment	Profit/(Loss) (₹ in lakhs)
A	225
B	25
C	(175)
D	(20)
E	(105)

Answer

As per AS 17 'Segment Reporting', a business segment or geographical segment should be identified as a reportable segment if:

Its segment results whether profit or loss is 10% or more of:

- The combined result of all segments in profit; i.e. ₹ 250 Lakhs or
- The combined result of all segments in loss; i.e. ₹ 300 Lakhs

Whichever is greater in absolute amount i.e. ₹ 300 Lakhs.

Operating Segment	Absolute amount of Profit or Loss (₹ in lakhs)	Reportable Segment (Yes or No)
A	225	Yes
B	25	No
C	175	Yes
D	20	No
E	105	Yes

On the basis of the profitability test (result criteria), segments **A, C and E are reportable segments** (since their results in absolute amount is 10% or more of₹ 300 lakhs i.e. 30 lakhs).

Question 2

ABC Limited has 5 segments namely A, B, C, D and E. The profit/loss of each segment for the year ended March 31, 20X2 is as follows:

Segment	Profit/(Loss) (₹ in crore)
A	780
B	1,500
C	(2,300)
D	(4,500)

E	6,000
Total	1,480

Identify the Reportable segments.

Answer

In compliance with AS 17, the segment profit/loss of respective segment will be compared with the greater of the following:

(i) All segments in profit, i.e., A, B and E - Total profit ₹ 8,280 crores.

(ii) All segments in loss, i.e., C and D - Total loss ₹ 6,800 crores.

Greater of the above - ₹ 8,280 crores.

Based on the above, reportable segments will be determined as follows:

Segment	Profit/(Loss)	Absolute Profit/Loss as a % of ₹ 8,280	Reportable Segment
A	780	9%	No
B	1,500	18%	Yes
C	(2,300)	28%	Yes
D	(4,500)	54%	Yes
E	6,000	72%	Yes
Total	**1,480**		

Question 3

M/s XYZ Ltd. has three segments namely X, Y, Z. The total Assets of the Company are ₹ 10 crores. Segment X has ₹ 2 crores, segment Y has ₹ 3 crores and segment Z has ₹ 5 crores. Deferred tax assets included in the assets of each segment are X - ₹ 0.50 crores, Y - ₹ 0.40 crores and Z - ₹ 0.30 crores. The accountant contends that all the three segments are reportable segments. Comment.

Answer

According to AS 17 "Segment Reporting", segment assets do not include income tax assets. Therefore, the revised total assets are ₹ 8.8 crores [₹ 10 crores – (₹ 0.5 + ₹ 0.4 + ₹ 0.3)]. Segment X holds total assets of ₹ 1.5 crores (₹ 2 crores – ₹ 0.5 crores); Segment Y holds ₹ 2.6 crores (₹ 3 crores – ₹ 0.4 crores); and Segment Z holds ₹ 4.7 crores (₹ 5 crores – ₹ 0.3 crores). Thus, all the three segments hold more than 10% of the total assets, all segments are reportable segments.

Question 4

Nathan Limited has three segments namely P, Q and R. The assets of the company are ₹ 15 crores. Segment P has ₹ 4 crores, Segment Q has ₹ 6 crores and Segment R has ₹ 5 crores. Deferred tax assets included in the assets of each segment are P - ₹ 1 crore, Q - ₹ 0.90 crores and R - ₹ 0.80 crores. The accountant contends all these three segments are reportable segments. Comment

Answer

According to AS 17 "Segment Reporting", segment assets do not include income tax assets.

Therefore, the revised total assets are ₹ 12.3 crores [₹ 15 – (₹ 1 + ₹ 0.9 + ₹ 0.8)].

Details of Segment wise assets:

Segment P holds total assets of ₹ 3 crores (₹ 4 crores – ₹ 1 crores);

Segment Q holds ₹ 5.1 crores (₹ 6 crores – ₹ 0.9 crores);

Segment R holds ₹ 4.2 crores (₹ 5 crores – ₹ 0.8 crores).

Thus, all the three segments hold more than 10% of the total assets, all segments are reportable segments.

Hence, the contention of the accountant that all three segments are reportable segments is **correct.**

Question 5

The Chief Accountant of Sports Ltd. gives the following data regarding its six segments:

₹ in lakhs

Particulars	M	N	O	P	Q	R	Total
Segment Assets	40	80	30	20	20	10	200
Segment Results	50	(190)	10	10	(10)	30	(100)
Segment Revenue	300	620	80	60	80	60	1,200

The Chief accountant is of the opinion that segments "M" and "N" alone should be reported. Is he justified in his view? Discuss.

Answer

As per **AS 17** 'Segment Reporting', a business segment or geographical segment should be identified as a reportable segment if:

Its revenue from sales to external customers and from other transactions with other segments is 10% or more of the total revenue- external and internal of all segments; or

Its segment result whether profit or loss is 10% or more of:

- The combined result of all segments in profit; or
- The combined result of all segments in loss, whichever is greater in absolute amount; or

Its segment assets are 10% or more of the total assets of all segments.

If the total external revenue attributable to reportable segments constitutes less than 75% of total enterprise revenue, additional segments should be identified as reportable segments even if they do not meet the 10% thresholds until atleast 75% of total enterprise revenue is included in reportable segments.

On the basis of turnover criteria segments **M and N are reportable** segments.

On the basis of the **result criteria**, segments **M, N and R** are reportable segments (since their results in absolute amount is 10% or more of ₹ 200 lakhs).

On the basis of **asset criteria**, **all segments except R** are reportable segments.

Since all the segments are covered in at least one of the above criteria, **all segments have to be reported** in accordance with Accounting Standard (AS) 17. Hence the opinion of chief **accountant is wrong**.

Question 6

Heavy Goods Ltd. has 6 segments namely L-Q (below). The total revenues (internal and external), profits or losses and assets are set out below: **(₹)**

Segment	Inter Segment Sales	External Sales	Profit/loss	Total assets
L	4,200	12,300	3,000	37,500
M	3,500	7,750	1,500	23,250
N	1,000	3,500	(1,500)	15,750
0	0	5,250	(750)	10,500
P	500	5,500	900	10,500
Q	1,200	1,050	600	5,250
	10,400	35,350	3,750	1,02,750

Heavy Goods Ltd. needs to determine how many reportable segments it has. You are required to advice Heavy Goods Ltd. as per the criteria defined in AS 17.

Answer

Quantitative Threshold Test:

Revenue Test:

Combined total sales of all the segment = ₹ 10,400 + ₹ 35,350 = ₹ 45,750

10% thresholds = ₹ 45,750 x 10% = 4,575

Profitability Test:

In the given situation, combined reported profit = ₹ 6,000 and combined reported loss (₹ 2,250). Hence, for 10% thresholds ₹ 6,000 will be considered.

10% thresholds = ₹ 6,000 x 10% = ₹ 600

Asset Test:

Combined total assets of all the segment = ₹ 1,02,750

10% thresholds = ₹ 1,02,750 x 10% = ₹ 10,275

Accordingly, quantitative thresholds are calculated below:

Segments	**L**	**M**	**N**	**O**	**P**	**Q**	**Reportable segments**
% segment sales to total sales	36.66%	24.59%	9.84%	11.48%	13.11%	4.92%	L, M, O, P
% segment profit to total profits	50%	25%	25%	12.5%	15%	10%	L, M, N, O, P, Q
% segment assets to total assets	36.50%	22.63%	15.33%	10.22%	10.22%	5.11%	L, M, N, O, P

Conclusion:

Segments L, M, O and P clearly satisfy the revenue and assets tests and they are separate reportable segments.

Segment N does not satisfy the revenue test, but it does satisfy the asset test and it is a reportable segment.

Segment Q does not satisfy the revenue or the assets test but is does satisfy the profits test. Therefore, Segment Q is also a reportable segment.

Hence all segments i.e. L, M, N, O, P and Q are reportable segments.

Preparation of Segment Report

Question 7

Calculate the segment results of a manufacturing organization from the following information:

Segments	A	B	C	Total
Directly attributed revenue	5,00,000	3,00,000	1,00,000	9,00,000
Enterprise revenue (allocated in 5:4:2 basis)				1,10,000
Revenue from transactions with other segments				
Transaction from B	1,00,000		50,000	1,50,000
Transaction from C	10,000	50,000		60,000
Transaction from A		25,000	1,00,000	1,25,000
Operating expenses	3,00,000	1,50,000	75,000	5,25,000
Enterprise expenses (allocated in 5:4:2 basis)				77,000
Expenses on transactions with other segments				
Transaction from B	75,000		30,000	
Transaction from C	6,000	40,000		
Transaction from A		18,000	82,000	

Answer

Computation of segment result:

Segments		A	B	C
Directly attributed revenue		5,00,000	3,00,000	1,00,000
Enterprise revenue (allocated in 5:4:2 basis)		50,000	40,000	20,000
Revenue from transactions with other segments				
Transaction from B		1,00,000	–	50,000
Transaction from C		10,000	50,000	–
Transaction from A		–	25,000	1,00,000
Total segment revenue	**(1)**	**6,60,000**	**4,15,000**	**2.70,000**
Operating expenses		3,00,000	1,50,000	75,000
Enterprise expenses (allocated in 5:4:2 basis)		35,000	28,000	14,000
Expenses on transactions with other segments				
Transaction from B		75,000	–	30,000
Transaction from C		6,000	40,000	–
Transaction from A		–	18,000	82,000
Total segment expenses	**(2)**	**4,16,000**	**2,36,000**	**2,01,000**
Segment result	**(1 – 2)**	**2,44,000**	**1,79,000**	**69,000**

Question 8

Prepare a segmental report for publication in Diversifiers Ltd. from the following details of the company's three divisions and the head office:

Particulars	₹ (000)
Forging Shop Division	
Sales to Bright Bar Division	4,575
Other Domestic Sales	90
Export Sales	6,135
	10,800
Bright Bar Division	
Sales to Fitting Division	45
Export Sales to Rwanda	300
	345
Fitting Division	
Export Sales to Maldives	270

₹ ('000)

Particulars	Head Office	Forging Shop Division	Bright Bar Division	Fitting Division
Pre-tax operating result		240	30	(12)
Head office cost reallocated		72	36	36
Interest costs		6	8	2
Fixed assets	75	300	60	180
Net current assets	72	180	60	135
Long-term liabilities	57	30	15	180

Answer

Diversifiers Ltd. Segmental Report **(₹ '000)**

Particulars	Divisions			Inter Segment Eliminations	Consolidated Total
	Forging shop	Bright Bar	Fitting		
Segment Revenue					
Sales:					
Domestic	90	–	–		90
Export	6,135	300	270		6,705
External Sales	6,225	300	270		6,795
Inter-Segment Sales	4,575	45	–	4,620	–
Total Revenue	**10,800**	**345**	**270**	**4,620**	**6,795**
Segment Result (Given)	240	30	(12)		258
Head Office Expenses					(144)
Operating Profit					114
Interest Expense					(16)
Profit Before Tax					**98**

Information in Relation to Assets and to Assets and Liabilities:					
Fixed Assets	300	60	180		540
Net Current Assets	180	60	135		375
Segment assets	480	120	315		915
Unallocated Corporate Assets (75 + 72)					147
Total assets					**1,062**
Segment liabilities	30	15	180		225
Unallocated corporate Liabilities					57
Total liabilities					**282**

Sales Revenue by Geographical Market **(₹ '000)**

Particulars	**Home Sales**	**Export Sales (by forging shop division)**	**Export to Rwanda**	**Export to Maldives**	**Consolidated Total**
External sales	90	6,135	300	270	6,795

CHAPTER - 18
AS 18: Related Party Disclosures

Identification of Related Party

Question 1

Consider a scenario wherein Mr. Robert holds 70% shares and voting rights in P Ltd.

Determine whether Andy (spouse of Mr. Robert) is a related party to P Ltd. under AS-18?

Answer

Yes – Andy is a related party to P Ltd., in view of the requirements of **AS-18**.

It may be recalled that under AS-18 '**relatives of individuals** owning an interest in the voting power of the reporting enterprise that gives them control or significant influence over the enterprise' are considered as related parties.

Question 2

Consider a scenario wherein:

- Mr. Robert is a Managing Director of P Ltd.
- Andy (spouse of Robert) received a remuneration of ₹ 5 lakhs from P Ltd. – for the services she rendered to P Ltd. for the period 1st April 20X1 through 30th June 20X1
- Andy left the services of P Ltd. on 1st July 20X1
- Consider 31st March 20X2 as the year-end date for P Ltd.

Whether Andy is to be identified as related party at the year-end date (31st March 20X2) for the purposes of AS-18?

Answer

Yes – This is because as per **AS-18**, parties are **considered** to be **related if at any time during** the reporting period one party has the **ability to control** the other party **or** exercise **significant influence** over the other party in making financial and/or operating decisions.

Hence Andy (being the spouse and relative of the KMP of P Ltd.) **needs to be reported** as related party at the year-end date (i.e. 31st March 20X2). This is because the remuneration Andy received from P Ltd. (for the period April 20X1 to 30th June 20X1) falls within the reporting year April 20X1 to March 20X2.

Question 3

Mr. Raj, a relative of key management personnel, received remuneration of ₹ 2,50,000 for his services in the company for the period from 1.4.20X1 to 30.6.20X1. On 1.7.20X1, he left the service of the company.

Should the relative be identified as at the closing date i.e. on 31.3.20X2 for the purposes of AS 18?

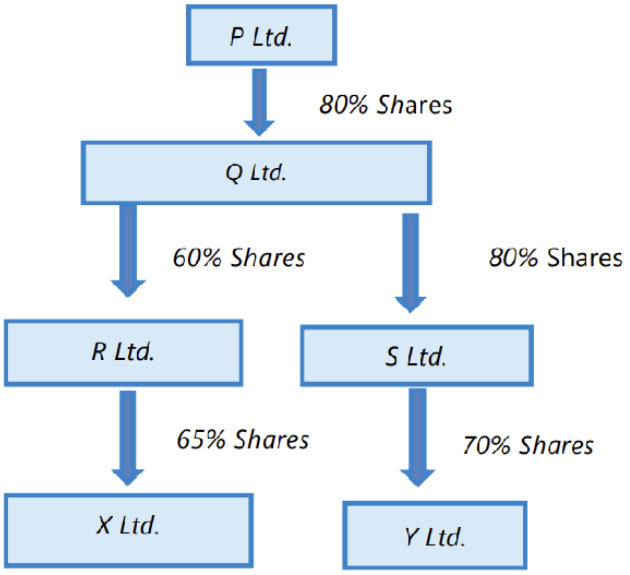
Answer

According to **AS 18** on 'Related Party Disclosures', parties are considered to be related if **at any time during the reporting period** one party has the **ability to control** the other party **or** exercise **significant influence**

over the other party in making financial and/or operating decisions. Hence Mr. Raj, a relative of key management personnel, **should be identified as related party** for disclosure in the financial statements for the year ended 31.3.20X2.

Question 4

Consider the following organization structure related to P Ltd.



Given the above structure: Identify related party relationships, if R Ltd. is the reporting enterprise.

Answer

The following table identifies the related party relationships for R Ltd. (being the reporting enterprise):

Party Name	Relationship under AS-18
P Ltd.	• P Ltd. has indirect control on R Ltd. (through Q Ltd.) • Hence, R Ltd. is **related** to P Ltd.
Q Ltd.	• Q Ltd. has direct control of R Ltd. • Hence, R Ltd. is **related** to Q Ltd.
S Ltd.	• R Ltd. and S Ltd. are under common control of Q Ltd. • Hence, R Ltd. is **related** to S Ltd.
X Ltd.	• X Ltd. is controlled by R Ltd. • Hence, R Ltd. is **related** to X Ltd.
Y Ltd.	• Y Ltd. is the sub-subsidiary of Q Ltd. • Both R Ltd. and Y Ltd. are under common control of Q Ltd. • Hence, R Ltd. is **related** to Y Ltd.

Question 5

Identify the related parties in the following case as per AS 18:

A Ltd. holds 51% of B Ltd.

B Ltd. holds 51% of O Ltd.

Z Ltd. holds 49% of O Ltd.

(Give your answer - Reporting Entity wise for A Ltd., B Ltd., O Ltd. and Z Ltd.)

Answer

In relation to Reporting enterprise - A Ltd.

- B Ltd. (subsidiary) is a related party
- O Ltd.(subsidiary) is a related party

In relation to Reporting enterprise - B Ltd.

- A Ltd. (holding company) is a related party
- O Ltd. (subsidiary) is a related party

In relation to Reporting enterprise - O Ltd.

- A Ltd. (ultimate holding company) is a related party
- B Ltd. (holding company) is a related party
- Z Ltd. (investor/investing party) is a related party (O Ltd being Associate of Z Ltd)

In relation to Reporting enterprise - Z Ltd.

- O Ltd. (Associate) is a related party

Question 6

You are required to identify the related parties in the following cases as per AS 18:

M Ltd. holds 61 % shares of S Ltd.

S Ltd. holds 51 % shares of F Ltd.

C Ltd. holds 49% shares of F Ltd.

(Give your answer - Reporting Entity wise for M Ltd., S Ltd., C Ltd. and F Ltd.)

Answer

Reporting Entity	Related Party
M Ltd.	S Ltd. (subsidiary) F Ltd.(subsidiary)
S Ltd.	M Ltd. (holding company) F Ltd. (subsidiary)
F Ltd.	M Ltd. (ultimate holding company) S Ltd. (holding company) C Ltd. (investor/investing party)
C Ltd.	F Ltd. (associate)

Question 7

The following information is provided for the year ended 31st March, 20X1:

(i) AX Limited holds 70% shares of BX Limited

(ii) BX Limited holds 30% shares of CX Limited

(iii) DX Limited holds 40% shares of in CX Limited

(iv) DX Limited holds 49% shares in EX Limited You are required to:

(a) Identity the related parties for the reporting entities – AX Limited, CX Limited and EX Limited.

(b) If DX Limited would have sold its investment in EX Limited on 1st October, 20X0, but goods were continued to be supplied by DX Limited to EX Limited throughout the year, will this scenario change your answer with respect to any of the reporting entity mentioned in point (a)?

Give reasons for your answer as per AS 18.

Answer

(a)

Reporting Entity	Related Party
AX Limited	**BX** Limited (AX Limited controls BX Limited directly as holding company **CX** Limited (AX Limited has significant influence in CX Limited indirectly through BX Limited)
CX Limited	**AX** Limited (AX Limited. has significant influence in CX Limited indirectly through BX Limited) **BX** Limited (BX Limited has significant influence in CX Limited) **DX** Limited (DX Limited holds 40% shares, means substantial interest of more than or equal to 20%, hence significant influence exists in this case)
EX Limited	**DX** Limited (DX Limited holds 49% shares, means substantial interest of more than or equal to 20%, hence significant influence exists in this case)

(b) **No,** DX would **still be a related party** of EX Limited EX Limited will report DX Limited as a related party as per AS 18. However, the transaction for the period in which the related party relationship exists will only be disclosed.

Question 8

Consider a scenario wherein:

- UK Bank holds 23% equity shares with voting rights in P Ltd.
- The bank has provided a loan of ₹ 20 million to P Ltd. at market interest rate
- As per the terms and conditions of the loan agreement, the bank has appointed one person as its nominee to the board of director of P Ltd. and any major transaction to be entered into by P Ltd. will require the consent of the Bank

Determine whether under AS-18 - UK Bank is a related party to P Ltd. (the reporting enterprise)?

Answer

In the instant case, the UK Bank holds 23% shares with voting rights in P Ltd. and hence is deemed to exercise significant influence over P Ltd.

The bank is also a provider of finance to P Ltd. (the reporting enterprise) and as per AS-18, parties like providers of finance are deemed not to be considered as a related party in the course of normal dealings with an enterprise by virtue only of those dealings. However, this exemption will not be available to UK Bank in this case – since it exercises significant influence over P Ltd. (by virtue of holding 23% shares with voting rights in P Ltd.)

Accordingly, for P Ltd. (the reporting enterprise), the **UK Bank is a related party** and it will be required to disclose the transactions with UK Bank in its financial statements.

Question 9

Consider a scenario wherein:

- A Ltd. has 60% voting right in B Ltd.
- A Ltd. also has 22% voting right in C Ltd.; and
- B Ltd. has 30% voting right in C Ltd.

Whether C Ltd. is to be treated under AS-18 as a party related to A Ltd.?

Answer

Yes – in relation to A Ltd. (the reporting enterprise), C Ltd. is a related party under AS-18. This is because A Ltd. indirectly controls C Ltd. In this case, A Ltd. (together with its subsidiary B Ltd.) controls more than one half of the voting rights of C Ltd.

Question 10

Consider the following organization structure related to UH Ltd. (the ultimate parent company of a Group), wherein UH Ltd. has made the following investments:

- Investment in two of the wholly owned subsidiaries, viz. Sub 1 and Sub 2
- Investment in JC 1, in which UH Ltd. has a joint control
- 28% investment in Ass 1 (and hence, Ass 1 is an associate of UH Ltd.)

Given the above structure: Identify related party relationships for each of the above entities under AS-18

Answer

The following table identifies the related party relationships for each of the entities in the Group:

Reporting enterprise	Related Party as per AS-18
UH Ltd.	All the four entities (viz. Sub 1, Sub 2, JC 1 and Ass 1)
Sub 1	Only two of the entities in the Group (viz. UH Ltd. and Sub 2)
Sub 2	Only two of the entities in the Group (viz. UH Ltd. and Sub 1)
JC 1	Only UH Ltd.
Ass 1	Only UH Ltd.

Question 11

Consider a scenario wherein:

- X Ltd. holds 28% voting right in Y Ltd. (and hence Y Ltd. is an associate of X Ltd.)
- Y Ltd. holds 32% voting right in Z Ltd. (and hence Z Ltd. is an associate of Y Ltd.)

In the above case, since Y Ltd. is an associate of X Ltd. – Y Ltd. is a related party to X Ltd.

Likewise, since Z Ltd. is an associate of Y Ltd. - Z Ltd. is a related party to Y Ltd.

Whether Z Ltd. is to be treated under AS-18 as a party related to X Ltd.?

Answer

No – in relation to X Ltd. (the reporting enterprise), **Z Ltd.** is a **not** a related party.

This is because as per the requirements of AS-18, '**associate of an associate' is not a related party**.

Question 12

A Limited has two Associates, B Limited and C Limited, and owns 25% of the voting power of B Limited and 30% of the voting power of C Limited. Would B Limited be considered a related party for the purpose of financial statements of C Limited?

Answer

Both B Limited and C Limited are 'associates' of A Limited. **Follow-associates cannot be regarded as a related parties** only by virtue of the relationship. AS 18 states that "enterprise that directly, or indirectly through one or more intermediaries, control, or are controlled by, or are under common control with, the reporting enterprise" are related parties. Further, it is given that "associates and joint ventures of the reporting enterprise and the investing party or venture in respect of which the reporting enterprise is an associate or a joint venturer" are also related parties. As B Limited is not an associate of C Limited, nor is it being controlled, directly or indirectly, by C Limited or is not so controlling C Limited, it **is not a related party** of C Limited.

Question 13

In respect of a key supplier who is dependent on the company for its existence and the company enjoys influence over the prices of this supplier (which may not be formally demonstrable), can the supplier and the company be considered as related parties?

Answer

The supplier and the company cannot be considered to be related parties merely because the latter is able to influence the transaction price between the parties. Paragraph 3 of AS 18 states that "enterprises that directly, or indirectly through one or more intermediaries, control, or are controlled by, or are under common control with, the reporting enterprise" are considered to be related party relationships. However, the conditions which define the existence of control, as follows, are not satisfied in the given example.

- Ownership, directly or indirectly, of more than one-half of the voting power of an enterprise, or
- Control of the composition of the board of directors in the case of a company or of the composition of the corresponding governing body in case of any other enterprise, or

- A substantial interest in voting power and the power to direct, by statue or agreement, the financial and/or operating policies of the enterprise".

Paragraph 10 of the standard defines significant influence as "participation in the financial and/or operating policy decisions of an enterprise, but not control of those policies". In the given example, although the supplier and the company have entered into a commercial transaction, the terms of which are influenced by the latter because of its better bargaining power in the specific market for such goods, it cannot be concluded that there is participation in the financial and/or operating policy decisions. Therefore, as the conditions specified by the Standard for being classified as a related party are not satisfied in the given example, the company **cannot be said to be related to the supplier.** This view is supported by paragraph 4 (b) of the Standard which states that "a single customer, supplier, franchiser, distributor, or general agent with whom an enterprise transacts a significant volume of business merely by virtue of the resulting economic dependence" would not be deemed to be related parties.

Question 14

Khushi Limited enter into an agreement with Mr. Happy for running a business for a fixed amount payable to the later every year. The contract states that the day-to-day management of the business will be handled by Mr. Happy, while all financial and operating policy decisions are taken by the Board of Directors of the Company. Mr. Happy does not own any voting power in Khushi Limited.

Answer

Mr. Happy will **not be considered as a related party** of Khushi Limited in view of provisions of AS 18 "Related Party Disclosures" which states, "individuals owning, directly or indirectly, an interest in the voting power of the reporting enterprise that gives them control or significant influence over the enterprise, and relatives of any such individual are related parties".

In the given case, in the absence of share ownership, Mr. Happy would not be considered to exercise significant influence on Khushi Limited, even though there is an agreement giving him the power to manage the company. Further, the fact that Mr. Happy does not have the ability to direct or instruct the board of directors does not qualify him as a key management personnel.

Related Party Disclosures

Question 15

Who are related parties under AS 18? What are the related party disclosure requirements?

Answer

Parties are considered to be related if at any time during the reporting period one party has the ability to **control** the other party or exercise **significant influence** over the other party in making financial and/or operating decisions.

If there have been transactions between related parties, during the existence of a related party relationship, the reporting enterprise should disclose the following:

(i) The **name** of the transacting related party;

(ii) A **description of the relationship** between the parties;

(iii) A description of the **nature of transactions**;

(iv) **Volume of the transactions** either as an amount or as an appropriate proportion;

(v) **Any other elements** of the related party transactions necessary for an understanding of the financial statements;

(vi) The amounts or appropriate proportions of outstanding items pertaining to related parties at the balance sheet date and **provisions for doubtful debts** due from such parties at that date;

(vii) Amounts **written off or written back** in the period in respect of debts due from or to related parties.

Question 16

Should the related parties be identified as at the reporting date (i.e. balance sheet date) for the purposes of AS-18? In disclosing transactions with related parties, are the transactions of the entire reporting period to be disclosed or only those for the period during which related party relationship exists?

Answer

As per the definition of related parties in **AS-18,** the existence of a related party relationship should be identified at all points during the year (and not only at the close of the financial year). However, AS 18 requires disclosure of transactions with these parties **only during the existence of the related party relationship.**

Question 17

X Ltd. sold goods to its associate company during the 1st quarter ended 30.6.20X1. After that, the related party relationship ceased to exist. However, goods were supplied as were supplied to any other ordinary customer. Decide whether transactions of the entire year have to be disclosed as related party transaction.

Answer

As per **AS 18**, transactions of X Ltd. with its associate company for the **first quarter ending 30.06.20X1** only are required to be **disclosed as related party transactions**. The **transactions** for the **period in which related party relationship did not exist would not be reported**.

Question 18

Consider a scenario wherein:

- P Ltd. hold 22% shares and voting rights in Q Ltd. (and hence Q Ltd. is an associate of P Ltd.)
- On 1st April 20X1, P Ltd. sold certain goods to Q Ltd. amounting to ₹ 5 lakhs
- On 30th June 20X1, P Ltd. sold its entire 22% stake in Q Ltd. (and hence the related party relationship ceased to exist after 30th June 20X1)
- However, P Ltd. continued supply goods to Q Ltd. subsequent to 30th June 20X1 (just like any other customer) and sold goods worth ₹ 15 lakhs during 9-month period ended 31st March 20X2
- Consider 31st March 20X2 as the year-end date for P Ltd.

Determine whether the transaction for the entire year (ending on 31st March 20X2) is required to be disclosed under AS-18 as related party transaction.

Answer

No – This is because as per AS-18, the disclosure requirements under the Standard relate only to the period **during related party relationship existed**.

Accordingly, only transactions between P Ltd and Q Ltd **till 30th June 20X1** (being sale of goods worth ₹. 5 lacs) are required to be reported/**disclosed under AS-18.**

Transactions entered into **after 30th June 20X1** are **not required** to be disclosed under AS-18.

Question 19

Narmada Ltd. sold goods for ₹ 90 lakhs to Ganga Ltd. during financial year ended 31.03.20X1. The Managing Director of Narmada Ltd. owns 100% shares of Ganga Ltd. The sales were made to Ganga Ltd. at normal selling prices by Narmada Ltd. The Chief accountant of Narmada Ltd. contends that these sales need not require a different treatment from the other sales made by the company and hence no disclosure is necessary as per the accounting standard. Is the Chief Accountant correct?

Answer

As per **AS 18** 'Related Party Disclosures', Enterprises over which key management personnel are able to exercise significant influence are related parties. This includes enterprises owned by directors or major shareholders of the reporting enterprise and enterprise that have a member of key management in common with the reporting enterprise.

In the given case, **Narmada Ltd. and Ganga Ltd. are related parties** and hence **disclosure of transaction** between them is **required irrespective of whether the transaction was done at normal selling price**.

Hence the contention of **Chief Accountant** of Narmada Ltd is **wrong**.

Question 20

Will transactions with related parties, for services provided/received free of cost, be required to be disclosed?

A Limited has a corporate communications department, which centralises the public relations function for the whole group of A Limited and its subsidiaries. No charges are, however, levied by A Limited on its subsidiaries and accordingly, these transactions are not given accounting recognition. Would these constitute related party transactions requiring disclosure under AS 18 in the standalone financial statements of A Limited?

Answer

These transactions would require disclosure under AS 18 in the standalone financial statements of A Limited. As per paragraph 10 of AS 18, a related party transaction is "a transfer of resources or obligations between related parties, regardless of whether or not a price is charged". In the given situation, there is a transfer of resources from A Limited to its subsidiaries, though no price is charged for the same. Hence, it will constitute as related party transaction and will require disclosure in the financial statements of A Ltd.

Question 21

Is remuneration paid to Board of Directors a related party transaction? Explain.

Answer

In case of a Company, the Managing Director, whole time director, manager and any person in accordance with whose directions or instructions the board of directors of the company is accustomed to act, are **usually considered Key Managerial Personnel (KMP).**

Persons who do not have the authority and responsibility for planning, directing and controlling the activities of the enterprise would not be KMP. Conversely, persons without any formal titles may be considered to be KMP, if they plan, direct and control the activities of the enterprise.

Further, as per Companies Act, 2013, a related party includes a director or his relative. The Act, defines a director as a director appointed to the Board of a Company.

Hence, remuneration paid to Board of Directors will be **considered as related party transaction**.

Question 22

ABC Limited is in the business of manufacturing textiles. It has certain commercial contracts with its customers and those customer contracts carry various clauses, imposing restriction on ABC Limited for disclosure of certain information. Accordingly, the company doesn't intend to provide related party disclosure under AS-18 in its ensuing financial statements. Is this correct?

Answer

As per **AS-18** stipulate that related party disclosure requirements under AS-18 **do not apply** in circumstances, where providing such disclosures would conflict with the reporting enterprise's duties of confidentiality, as specifically required in terms of a statute or by any regulator or similar competent authority.

In case, where (1) a statute or (2) a regulator or (3) a similar competent authority governing an enterprise prohibit the enterprise to disclose certain information, which is required to be disclosed as per AS 18, disclosure of such information is **not warranted**. For example, banks are obliged by law to maintain confidentiality in respect of their customers transactions and AS-18 would not override the obligation to preserve the confidentiality of customers dealings.

However, this exemption is **not available** in respect of confidentiality provisions in a commercial contract between two enterprises - where confidentiality is not specifically required in terms of (1) a statute or (2) by any regulator or (3) similar competent authority.

Therefore, in the given case AS-18 related party **disclosures would have to be made** by ABC Limited in its ensuing financial statements.

Mixed Topics

Question 23

Answer the following with respect to AS-18:

(i) ABC Ltd. sold goods of ₹ 2,00,000 to its associate company for the 1st quarter ending 30.06.20X1. After that the related party relationship ceased to exist. However, goods were supplied to any other ordinary customer. Decide whether transactions of the entire year have to be disclosed as related party transaction.

(ii) If the majority of directors of Arjun Ltd. constitute the majority of the Board of another Company Bheem

Ltd. in their individual capacity as professionals (and not by virtue of their being Directors in Arjun Ltd.). Are both the companies related?

(iii) Asha Ltd. sells all the manufactured furniture of ₹ 1,00,00,000 to Sasha Ltd, as per agreement. Sasha Ltd. is the only customer to Asha Ltd. In the financial statements, Asha Ltd. wants to present Sasha company as a related party. Comment on the disclosure requirement.

Answer

(i) As per AS 18, parties are considered to be related if any time during the reporting period one party has the ability to control the other party or exercise significant influence over the other party. Transactions of ABC Ltd. with its associate company for the **first quarter ending 30.06.20X1** only are required to be **disclosed as related party transactions** as the company has the ability to exercise significant influence only till 30.6.20X1.

The transactions for the period in which related party relationship did not exist need not be reported.

(ii) In the given case, Arjun Ltd. cannot be said to control the composition of board of directors of Bheem Ltd. as the directors have been appointed in their individual capacity as professionals and not by virtue of their being directors in Arjun Ltd.

Hence, it **cannot be concluded** that the companies are **related** merely because the majority of the directors of one company became the majority of the directors of the second in their individual capacity as professionals.

(iii) In the context of AS 18, a single customer, supplier, franchiser, distributor, or general agent with whom an enterprise transacts a significant volume of business cannot be construed as Related Party Relationship merely by virtue of the resulting economic dependence. There is an economic dependence between the companies but no one controls or exercise significant influence on the other.

In the given case, Asha Ltd. need **not report** Sasha Company as its **related party** in its financial statements.

CHAPTER - 19
AS 19: Leases

Accounting of Operating Lease

Question 1

A machine was given on 3 years operating lease by a dealer of the machine for equal annual lease rentals to yield 30% profit margin on cost ₹ 1,50,000. Economic life of the machine is 5 years and output from the machine are estimated as 40,000 units, 50,000 units, 60,000 units, 80,000 units and 70,000 units consecutively for 5 years. Straight line depreciation in proportion of output is considered appropriate. Compute the following:

(i) Annual Lease Rent

(ii) Lease Rent income to be recognized in each operating year and

(iii) Depreciation for 3 years of lease

Answer

(i) Annual lease rent

Total lease rent

= 130% of ₹ 1,50,000 x (Output during lease period/Total output)

= 130% of ₹ 1,50,000 x [(40,000 + 50,000 + 60,000)/(40,000 + 50,000 + 60,000 + 80,000 + 70,000)]

= 1,95,000 x (1,50,000 units/3,00,000 units) = ₹ 97,500

Annual lease rent = ₹ 97,500/3 = ₹ 32,500

(ii) Lease rent Income to be recognized in each operating year

Total lease rent should be recognised as income in proportion of output during lease period, i.e. in the proportion of 40:50:60.

Hence income recognised in years 1, 2 and 3 will be as:

Year 1 ₹ 26,000,

Year 2 ₹ 32,500 and

Year 3 ₹ 39,000.

(iii) Depreciation for three years of lease

Since depreciation in proportion of output is considered appropriate, the depreciable amount ₹ 1,50,000 should be allocated over useful life 5 years in proportion of output, i.e. in proportion of 40:50:60:80:70.

Depreciation for year 1 is ₹ 20,000, year 2 = 25,000 and year 3 = 30,000.

Accounting of Finance Lease in Lessee's Books

Question 2

S. Square Private Limited has taken machinery on finance lease from S.K. Ltd. The information is as under:

Lease term = 4 years

Fair value at inception of lease = ₹ 20,00,000

Lease rent = ₹ 6,25,000 p.a. at the end of year

Guaranteed residual value = ₹ 1,25,000

Expected residual value = ₹ 3,75,000

Implicit interest rate = 15%

Discounted rates for 1st year, 2nd year, 3rd year and 4th year are 0.8696, 0.7561, 0.6575 and 0.5718 respectively.

Calculate the value of the lease liability as per AS-19.

Answer

According to para 11 of **AS 19 "Leases",** the **lessee** should **recognise** the lease as an **asset and a liability** at an amount equal to the **lower of the fair value** of the leased asset at the inception of the finance lease and the **present value of the minimum lease payments** from the standpoint of the lessee.

In calculating the present value of the minimum lease payments the discount rate is the interest rate implicit in the lease. Present value of minimum lease payments will be calculated as follows:

Year	Minimum Lease Payment (₹)	PV factor @15%	Present value (₹)
1	6,25,000	0.8696	5,43,500
2	6,25,000	0.7561	4,72,563
3	6,25,000	0.6575	4,10,937
4	(6,25,000 + 1,25,000) = 7,50,000	0.5718	4,28,850
Total	**26,25,000**		**18,55,850**

Present value of **minimum lease payments ₹ 18,55,850 is less than fair value** at the inception of lease i.e. ₹ 20,00,000, therefore, the **asset and corresponding lease liability should be recognised at ₹ 18,55,850** as per AS 19.

Question 3

You are required to give the necessary journal entry at the inception of lease to record the asset taken on finance lease in books of lessee from the following information:

Lease period = 5 years

Annual lease rents = ₹ 50,000 at the end of each year

Guaranteed residual value = ₹ 25,000

Fair Value at the inception (beginning) of lease = ₹ 2,00,000

Interest rate implicit on lease is = 12.6% (Discounted rates for year 1 to 5 are .890, .790, .700, .622 and .552 respectively).

Answer

Present value of minimum lease payment is computed below:

Year	MLP (₹)	DF (12.6%)	PV (₹)
1	50,000	0.890	44,500
2	50,000	0.790	39,500
3	50,000	0.700	35,000

4	50,000	0.622	31,100
5	50,000	0.552	27,600
5	25,000	0.552	13,800
			1,91,500

Present value of minimum lease payment = ₹ 1,91,500

Fair value of leased asset = ₹ 2,00,000

As per **AS 19,** on the date of inception of Lease, Lessee should **show it as an asset and corresponding liability** at **lower of Fair value** of leased asset at the inception of the lease and **present value of minimum lease payments** from the standpoint of the lessee. The accounting entry at the inception of lease to record the asset taken on finance lease in books of lessee is suggested below:

Particulars		**Debit (₹)**	**Credit (₹)**
Asset A/c	Dr.	1,91,500	
To Lessor (Lease Liability) A/c			1,91,500
(Being recognition of finance lease as asset and liability)			

Question 4

Lessee Ltd. took a machine on lease from Lessor Ltd., the fair value being ₹ 7,00,000.

The economic life of machine as well as the lease term is 3 years. At the end of each year Lessee Ltd. pays ₹ 3,00,000. The Lessee has guaranteed a residual value of ₹ 22,000 on expiry of the lease to the Lessor. However, Lessor Ltd., estimates that the residual value of the machinery will be only ₹ 15,000. The implicit rate of return is 15% p.a. and present value factors at 15% are 0.869, 0.756 and 0.657 at the end of first, second and third years respectively.

Calculate the value of machinery to be considered by Lessee Ltd. and the finance charges in each year.

Answer

As per para 11 of **AS 19 "Leases",** the **lessee** should **recognize** the lease as an **asset and a liability** at the inception of a finance lease. Such recognition should be at an amount equal to the **fair value** of the leased asset at the inception of lease. However, **if the fair value of the leased asset exceeds the present value of minimum lease payment** from the standpoint of the lessee, the amount **recorded as an asset and liability should be the present value of minimum lease payments** from the standpoint of the lessee.

Computation of Value of machinery:

Present value of minimum lease payment = ₹ 6,99,054 (See working note below)

Fair value of leased asset = ₹ 7,00,000

Therefore, the recognition will be at the lower of the two i.e. 6,99,054

Working Note - Present value of minimum lease payments:

Annual lease rental x PVIF + Present value of guaranteed residual value

= ₹ 3,00,000 x (0.869 + 0.756 + 0.657) + ₹ 22,000 x 0.657

= ₹ 6,84,600 + ₹ 14,454

= 6,99,054

Computation of finance charges:

Year	Finance charge	Payment	Reduction in o/s Liability	Outstanding liability
1st Year beginning	–	–	–	6,99,054
End of 1st year	1,04,858	3,00,000	1,95,142	5,03,912
End of 2nd year	75,587	3,00,000	2,24,413	2,79,499
End of 3rd year	41,925	3,00,000	2,58,075	21,424

Accounting of Finance Lease in Lessor's Books

Question 5

Prakash Limited leased a machine to Badal Limited on the following terms:

		(₹ In lakhs)
(i)	Fair value of the machine	28.3
(ii)	Lease term	5 years
(iii)	Lease rental per annum	8.00
(iv)	Guaranteed residual value	1.60
(v)	Expected residual value	3.00
(vi)	Internal rate of return	15%

Discounted rates for 1st year to 5th year are 0.8696, 0.7561, 0.6575, 0.5718, and 0.4972 respectively.

Ascertain Unearned Finance Income.

Answer

As per **AS 19** on Leases, **unearned finance income** is the difference between (a) the **gross investment** in the lease and (b) the **present value of minimum lease payments** under a finance lease from the standpoint of the lessor; and any unguaranteed residual value accruing to the lessor, at the interest rate implicit in the lease.

Where:

(a) **Gross investment** in the lease is the aggregate of (i) minimum lease payments from the stand point of the lessor and (ii) any unguaranteed residual value accruing to the lessor.

Gross investment = Minimum lease payments + Unguaranteed residual value (URV)

= [Total lease rent + Guaranteed residual value (GRV)] + URV

= [(₹ 8,00,000 x 5 years) + ₹ 1,60,000] + ₹ 1,40,000

= ₹ 43,00,000

(b) Table showing present value of (i) Minimum lease payments (MLP) and (ii) Unguaranteed residual value

Year	MLP inclusive of URV (₹)	PV Factors @ 15%	Present Value (₹)
1	8,00,000	0.8696	6,95,680
2	8,00,000	0.7561	6,04,880
3	8,00,000	0.6575	5,26,000
4	8,00,000	0.5718	4,57,440
5	8,00,000	0.4972	3,97,760

(GRV)	1,60,000	0.4972	79,552
	41,60,000		27,61,312
(URV)	1,40,000	0.4972	69,608
	43,00,000		**28,30,920**

Unearned Finance Income = ₹ 43,00,000 – ₹ 28,30,920 = **₹ 14,69,080**

Question 6

J Limited availed an equipment on lease from K Limited. The conditions of the lease terms are as under:

(i) Lease starts from 1st April, 20X1 for a period of 4 Years and useful life of the equipment is 6 years. Both the cost and fair value of equipment are ₹ 12,50,000.

(ii) The equipment reverts back to the lessor on termination of the lease.

(iii) The unguaranteed residual value is estimated at ₹ 1,20,000 at the end of the financial year 20X4-20X5.

(iv) The amount will be paid in 4 equal instalments at the end of each year.

(v) Consider IRR = 8%.

(vi) The present value of ₹ 1 at the end of 4th year at 8% of interest is ₹ 0.735.

(vii) The present value of annuity of ₹ 1 due at the end of 4th year at 8% IRR is ₹ 3.312

State whether this lease is operating lease or Finance lease (by applying two deterministic parameters). Also calculate unearned finance Income.

Answer

Computation of annual lease payment:

Particulars	**₹**
Cost of equipment	12,50,000.00
Unguaranteed residual value	1,20,000.00
Present value of unguaranteed residual value (₹ 1,20,000 x 0.735)	88,200.00
Present value of lease payments (₹ 12,50,000 – ₹ 88,200)	11,61,800.00
Present value of annuity for four years is 3.312	
Annual lease payment [11,61,800/3.312]	**3,50,785.02**

Classification of lease:

Parameter 1:

The present value of the lease payment i.e. ₹ 11,61,800 which equals **92.94%** of the fair market value i.e. ₹ 12,50,000.

The present value of minimum lease payments **substantially covers the fair value** of the leased asset.

Parameter 2:

The lease term (i.e. 4 years) **covers the major part of the life of asset** (i.e. 6 years).

Therefore, it **constitutes a finance lease**.

Computation of Unearned Finance Income:

Particulars	**₹**

Total lease payments (₹ 3,50,785.02 x 4)	14,03,140.08
Add: Unguaranteed residual value	1,20,000.00
Gross investment in the lease	15,23,140.08
Less: PV of lease payments and residual value i.e. Net Investment (₹ 88,200 + ₹ 11,61,800)	(12,50,000.00)
Unearned finance income	**2,73,140.08**

Accounting of Sale and Lease Back

Question 7

On 1st April, 20X1, Mansi Ltd. sold a plant for ₹ 8,52,800. The carrying amount of the plant on that date was ₹ 1,80,000. The sale was a part of the package under which Akash Ltd. leased the asset to Mansi Ltd. for eight years term.

The economic life of the asset is estimated as 8 years. The minimum lease rents payable by the lessee has fixed at ₹ 1,60,000 payable annually beginning from 31st March, 20X2.

The incremental borrowing interest rate of Mansi Ltd. is estimated at 10% p.a.

Calculate the net effect on the Statement of profit and loss in the books of Mansi Ltd.

Answer

Net effect on the Statement of Profit and Loss in the year of sale in the books of Lessee (Mansi Ltd.)

For calculation of net effect on the statement of profit and loss on sale of equipment, it has to be judged whether lease is an operating lease or finance lease.

The **lease term is for 8 years** which **covers the entire economic life of the equipment**. At the inception of the lease, the present value of the **minimum lease payments (MLP) is ₹ 8,53,600** [₹ 1,60,000 x 5.335 (Annuity factor of ₹ 1 @10% for 8 years)] and amounts to at least **substantially all of the fair value** (sale price i.e. ₹ 8,52,800) of the leased equipment. Thus, lease is a **finance lease**.

As per para 48 of AS 19 "Leases", if a sale and leaseback transaction results in a **finance lease, profit of ₹ 6,72,800** (Sale value ₹ 8,52,800 less carrying amount ₹ 1,80,000) will **not be recognized** as income in the **year of sale** in the books of lessee i.e. Mansi Ltd. It **should be deferred and amortised over the lease term** in proportion to the depreciation of the leased asset.

Therefore, assuming that **depreciation** is charged on straight line basis, Mansi Ltd. will recognize depreciation of **₹ 1,06,600 per annum** for 8 years (₹ 8,52,800/8) and **amortise profit** of ₹ 6,72,800 over the lease term of 8 years, i.e. **₹ 84,100 p.a.**

The **net effect is a debit** of (₹ 1,06,600 – 84,100) **₹ 22,500 p.a**. to the Statement of Profit and Loss, for 8 years as covered under the lease term.

Question 8

X Ltd. sold machinery having WDV of ₹ 300 lakhs to Y Ltd. for ₹ 400 lakhs and the same machinery was leased back by Y Ltd. to X Ltd. The lease back arrangement is operating lease.

Give your comments in the following situations:

(i) Sale price of ₹ 400 lakhs is equal to fair value.

(ii) Fair value is ₹ 450 lakhs.

(iii) Fair value is ₹ 350 lakhs and the sale price is ₹ 250 lakhs.

(iv) Fair value is ₹ 300 lakhs and sale price is ₹ 400 lakhs.

(v) Fair value is ₹ 250 lakhs and sale price is ₹ 290 lakhs.

Answer

Accounting Treatment:

S. No.	Particulars	Accounting Treatment
(i)	When sale price of ₹ 400 lakhs is equal to fair value	X Ltd. should **immediately** recognize the **profit of ₹ 100** lakhs (i.e. 400 – 300) in its books.
(ii)	When fair value is ₹ 450 lakhs	**Profit of ₹ 100** lakhs should be **immediately** recognized by X Ltd.
(iii)	When fair value of leased machinery is ₹ 350 lakhs & sales price is ₹ 250 lakhs	Then, **loss of ₹ 50 lakhs** (300 – 250) to be **immediately recognized** by X Ltd. in its books provided loss is not compensated by future lease payment.
(iv)	When fair value is ₹ 300 lakhs & sales price is ₹ 400 lakhs	Then, **profit of ₹ 100** lakhs is to be deferred and **amortized over the lease period**.
(v)	When fair value is ₹ 250 lakhs & sales price is ₹ 290 lakhs	Then, the **loss of ₹ 50 lakhs** (300 – 250) to be **immediately** recognized by X Ltd. in its books and **profit of ₹ 40 lakhs** (290 – 250) should be **amortized/deferred over lease period**.

Question 9

A Ltd. sold machinery having WDV of ₹ 40 lakhs to B Ltd. for ₹ 50 lakhs and the same machinery was leased back by B Ltd. to A Ltd. The lease back is operating lease. Comment if

(a) Sale price of ₹ 50 lakhs is equal to fair value.

(b) Fair value is ₹ 60 lakhs.

(c) Fair value is ₹ 45 lakhs and sale price is ₹ 38 lakhs.

(d) Fair value is ₹ 40 lakhs and sale price is ₹ 50 lakhs.

(e) Fair value is ₹ 46 lakhs and sale price is ₹ 50 lakhs

(f) Fair value is ₹ 35 lakhs and sale price is ₹ 39 lakhs.

Answer

Following will be the treatment in the given cases:

(a) When sales price of ₹ 50 lakhs is equal to fair value, A Ltd. should **immediately** recognise the **profit of ₹ 10 lakhs** (i.e. 50 – 40) in its books.

(b) When fair value is ₹ 60 lakhs then also **profit of ₹ 10 lakhs** should be **immediately** recognised by A Ltd.

(c) When fair value of leased machinery is ₹ 45 lakhs & sales price is ₹ 38 lakhs, then **loss of ₹ 2 lakhs** (40 – 38) to be **immediately** recognised by A Ltd. in its books provided loss is not compensated by future lease payment, otherwise defer and amortise the loss.

(d) When fair value is ₹ 40 lakhs & sales price is ₹ 50 lakhs then, **profit of ₹ 10 lakhs** is to be deferred and amortised **over the lease period**.

(e) When fair value is ₹ 46 lakhs & sales price is ₹ 50 lakhs, profit of **₹ 6 lakhs** (46 – 40) to be **immediately** recognised in its books and balance profit of **₹ 4 lakhs** (50 – 46) is to be amortised/deferred **over lease period.**

(f) When fair value is ₹ 35 lakhs & sales price is ₹ 39 lakhs, then the **loss of ₹ 5 lakhs** (40 – 35) to be **immediately** recognised by A Ltd. in its books and **profit of ₹ 4 lakhs** (39 – 35) should be amortised/deferred **over lease period.**

Determination of Operating Lease or Finance Lease

Question 10

Classify the following into either operating or finance lease:

(i) Lessee has option to purchase the asset at lower than fair value, at the end of lease term;

(ii) Economic life of the asset is 7 years, lease term is 6 years, but asset is not acquired at the end of the lease term;

(iii) Economic life of the asset is 6 years, lease term is 2 years, but the asset is of special nature and has been procured only for use of the lessee;

(iv) Present value (PV) of Minimum lease payment (MLP) "X" = Fair value of the asset is "Y".

(v) Lease payments should be recognized as an expense in the statement of Profit & Loss of a lessee

Answer

(i) If it becomes certain at the inception of lease itself that the option will be exercised by the lessee, it is a **Finance Lease**.

(ii) The lease will be classified as a **finance lease**, since a substantial portion of the life of the asset is covered by the lease term.

(iii) Since the asset is procured only for the use of lessee, it is a **finance lease**.

(iv) The lease is a **finance lease** if X = Y, or where X substantially equals Y.

(v) It is an **operating lease** under which lease payments are recognized as expense in the profit and loss account of lessee to have better matching between cost and revenue.

Question 11

Sun Limited wishes to obtain a machine costing ₹ 30 lakhs by way of lease. The effective life of the machine is 14 years, but the company requires it only for the first 3 years. It enters into an agreement with Star Ltd., for a lease rental for ₹ 3 lakhs p.a. payable in arrears and the implicit rate of interest is 15%. The chief accountant of Sun Limited is not sure about the treatment of these lease rentals and seeks your advice. (Use annuity factor at @ 15% for 3 years as 2.28)

Answer

As per **AS 19** 'leases', a lease will be classified as **finance lease** if at the inception of the lease, the **present value of minimum lease payment** amounts to at least **substantially all of the fair value** of leased asset. In the given case, the implicit rate of interest is given at 15%. The present value of minimum lease payments at 15% using PV- Annuity Factor can be computed as:

Annuity Factor (Year 1 to Year 3)	2.28
Present Value of minimum lease payments (₹ 3 lakhs each year)	₹ 6.84 lakhs

Thus, present value of **minimum lease payments is ₹ 6.84 lakhs** and the fair value of the **machine is ₹ 30 lakhs**. In a **finance lease**, the **lease term** should be for a **major part of the economic life** of the asset even if title is not transferred. However, in the given case, the effective **useful life of the machine is 14 years** while the **lease is only for three years**. **Therefore**, lease agreement is an **operating lease**. **Lease payments** under an operating lease should be **recognized as an expense** in the statement of profit and loss on a **straight-line basis over the lease term** unless another systematic basis is more representative of the time pattern of the user's benefit.

CHAPTER - 20
AS 20: Earnings Per Share

Calculation of Basic EPS

Question 1

Date	Particulars			Balance
1st January	Balance at beginning of year	1,800	–	1,800
31st May	Issue of shares for cash	600	–	2,400
1st November	Buy Back of shares	–	300	2,100

Calculate Weighted Number of Shares.

Answer

Computation of Weighted Average:

(1,800 x 12/12) + (600 x 7/12) – (300 x 2/12) = 2,100 shares

Question 2

Date	Particulars	No. of Shares	Face Value	Paid up Value
1st January	Balance at beginning of year	1,800	₹ 10	₹ 10
31st October	Issue of Shares	600	₹ 10	₹ 5

Calculate Weighted Number of Shares.

Answer

Assuming that partly paid shares are entitled to participate in the dividend to the extent of amount paid, number of partly paid equity shares would be taken as 300 for the purpose of calculation of earnings per share.

Computation of weighted average would be as follows: (1,800 x 12/12) + (600 x 2/12 x 5/10) **= 1,850 shares**

Question 3

On 1st April, 20X1 a company had 6,00,000 equity shares of ₹ 10 each (₹ 5 paid up by all shareholders). On 1st September, 20X1 the remaining ₹ 5 was called up and paid by all shareholders except one shareholder having 60,000 equity shares. The net profit for the year ended 31st March, 20X2 was ₹ 21,96,000 after considering dividend on preference shares of ₹ 3,40,000.

You are required to compute Basic EPS for the year ended 31st March, 20X2 as per Accounting Standard 20 "Earnings Per Share".

Answer

$$\text{Basic Earnings per share (EPS)} = \frac{\text{Net profit attributable to equity shareholders}}{\text{Weighted average number of equity shares outstanding during the year}}$$

$$= \frac{21,96,000}{4,57,500 \text{ Shares (as per working note)}} = ₹ 4.80 \text{ per share}$$

Working Note:

Calculation of weighted average number of equity shares

As per AS 20 'Earnings Per Share', partly paid equity shares are treated as a fraction of equity share to the extent that they were entitled to participate in dividend relative to a fully paid equity share during the reporting period. Assuming that the partly paid shares are entitled to participate in the dividend to the extent of amount paid, weighted average number of shares will be calculated as follows:

Date	No. of equity Shares	Amount paid per share	Weighted average no. of equity shares
1.4.20X1	6,00,000	5	6,00,000 x 5/10 x 5/12 = 1,25,000
1.9.20X1	5,40,000	10	5,40,000 x 7/12 = 3,15,000
1.9.20X1	60,000	5	60,000 x 5/10 x 7/12 = 17,500
Total weighted average equity shares			**4,57,500**

Calculation of Basic EPS in Case of Bonus Issue

Question 4

Net profit for the year 20X1 ₹ 18,00,000

Net profit for the year 20X2 ₹ 60,00,000

No. of equity shares outstanding until 30th September 20X2 20,00,000

Bonus issue 1st October 20X2 was 2 equity shares for each equity share outstanding at 30th September, 20X2

Calculate Basic Earnings Per Share.

Answer

No. of Bonus Issue 20,00,000 x 2 = 40,00,000 shares

Earnings per share for the year 20X2 = $\frac{₹\ 60,00,000}{(20,00,000 + 40,00,000)}$ = ₹ 1.00

Adjusted earnings per share for the year 20X1 = $\frac{₹\ 18,00,000}{(20,00,000 + 40,00,000)}$ = ₹ 0.30

Since the bonus issue is an issue without consideration, the issue is treated as if it had occurred prior to the beginning of the year 20X1, the earliest period reported.

Question 5

NAT, a listed entity, as on 1st April, 20X1 had the following capital structure:

Particulars	₹
10,00,000 Equity Shares having face value of ₹ 1 each	10,00,000
10,00,000 8% Preference Shares having face value of ₹ 10 each	1,00,00,000

During the year 20X1-20X2, the company had profit after tax of ₹ 90,00,000.

On 1st January, 20X2, NAT made a bonus issue of one equity share for every 2 equity shares outstanding as at 31st December, 20X1.

On 1st January, 20X2, NAT issued 2,00,000 equity shares of ₹ 1 each at their full market price of ₹ 7.60 per share.

NAT's shares were trading at ₹ 8.05 per share on 31st March, 20X2.

Further it has been provided that the basic earnings per share for the year ended 31st March, 20X1 was previously reported at ₹ 62.30.

You are required to:

(i) Calculate the basic earnings per share to be reported in the financial statements of NAT for the year ended 31st March, 20X2 including the comparative figure, in accordance with AS-20 Earnings Per Share.

(ii) Explain why the bonus issue of shares and the shares issue at full market price are treated differently in the calculation of the basic earnings per share?

Answer

(i) Computation of Basic Earnings per share for the year ended 31st March, 20X2: (including the comparative figure)

Earnings for the year ended 31st March, 20X1 = EPS x Number of shares outstanding during 20X0-20X1

= ₹ 62.30 x 10,00,000 equity shares

= ₹ 6,23,00,000

Adjusted/Restated Earnings per share for the year ended 31st March 20X1: (after taking into consideration bonus issue)

Adjusted/Restated Basic EPS = Earnings for year 20X0-20X1/(Total outstanding shares + Bonus issue)

= ₹ 6,23,00,000/(10,00,000 + 5,00,000)

= ₹ 6,23,00,000/15,00,000 = ₹ 41.53 per share

Computation of Basic EPS for the year 20X1-20X2:

Basic EPS = (Total Earnings – Preference Shares Dividend)/(Total shares outstanding at the beginning + Bonus issue + weighted average of the shares issued in January, 20X2)

= (₹ 90,00,000 – ₹ (1,00,00,000 x 8%))/(10,00,000 + 5,00,000 + (2,00,000 x 3/12))

= ₹ 82,00,000/15,50,000 shares = ₹ 5.29 per share

(ii) In case of a **bonus issue, equity shares are issued to existing shareholders for no additional consideration**. Therefore, the number of equity shares outstanding is increased without an increase in resources. Since the bonus issue is an issue without consideration, the issue is **treated as if it had occurred prior to the beginning of the year 20X1**, the earliest period reported.

However, the **share issued at full market price does not carry any bonus element** and usually results in a proportionate change in the resources available to the enterprise. Therefore, it is **taken into consideration from the time it has been issued** i.e. the time- weighting factor is considered based on the specific shares outstanding as a proportion of the total number of days in the period.

Calculation of Basic EPS in Case of Rights Issue

Question 6

Net profit for the year 20X1	₹ 11,00,000
Net profit for the year 20X2	₹ 15,00,000
No. of shares outstanding prior to rights issue	5,00,000 shares

Rights issue price ₹ 15.00

Last date to exercise rights 1st March 20X2

Rights issue is one new share for each five outstanding (i.e. 1,00,000 new shares)

Fair value of one equity share immediately prior to exercise of rights on 1st March 20X2 was ₹ 21.00. Compute Basic Earnings Per Share.

Answer

Theoretical ex-rights fair value per share:

$$= \frac{\text{Fair value of shares immediately prior to exercise of rights + Total amount received from exercise}}{\text{Number of shares outstanding prior to exercise + Number of shares issued in the exercise}}$$

$$= \frac{(₹\ 21.00 \times 5{,}00{,}000 \text{ shares}) + (₹\ 15.00 \times 1{,}00{,}000 \text{ Shares})}{5{,}00{,}000 \text{ Shares} + 1{,}00{,}000 \text{ Shares}}$$

Theoretical ex-rights fair value per share = ₹ 20.00

Computation of adjustment factor:

$$= \frac{\text{Fair value per share prior to exercise of rights}}{\text{Theoretical ex-rights value per share}}$$

$$= \frac{₹\ (21.00)}{₹\ (20.00)} = 1.05$$

Computation of earnings per share:

EPS for the year 20X1 as originally reported: ₹ 11,00,000/5,00,000 shares = ₹ 2.20

EPS for the year 20X1 restated for rights issue: ₹ 11,00,000/ (5,00,000 shares x 1.05) = ₹ 2.10

EPS for year 20X2 incl. effects of rights issue: (5,00,000 x 1.05 x 2/12) + (6,00,000 x 10/12) = 5,87,500 shares

EPS = 15,00,000/5,87,500 = ₹ 2.55

Question 7

X Ltd. supplied the following information. You are required to compute the basic earnings per share:

(Accounting year 1.1.20X1-31.12.20X1)

Net Profit	: Year 20X1: ₹ 20,00,000
	: Year 20X2: ₹ 30,00,000
No. of shares outstanding prior to Right Issue	: 10,00,000 shares
Right Issue	: One new share for each four outstanding i.e., 2,50,000 shares
	Right Issue price - ₹ 20
	Last date of exercise right - 31.3.20X2
Fair rate of one Equity shares immediately prior to exercise of rights on 31.3.20X2	: ₹ 25

Answer

Computation of Basic Earnings Per Share

	Year 20X1 (₹)	Year 20X2 (₹)
EPS for the year 20X1 as originally reported		
[Net profit attributable to equity shareholders/Weighted average number of equity shares outstanding during the year]		
= (₹ 20,00,000/10,00,000 shares)	2.00	
EPS for the year 20X1 restated for rights issue		
= [₹ 20,00,000/{10,00,000 shares x 1.04 (WN 2)}]	1.91(approx.)	
EPS for the year 20X2 including effects of rights issue		
= ₹ 30,00,000/[(10,00,000 share x 1.04 x 3/12) + (12,50,000 share x 9/12)]		
= ₹ 30,00,000/11,97,500 shares		2.51(approx.)

Working Notes:

1. Computation of theoretical ex-rights fair value per share

$$= \frac{\text{Fair value of all outstanding shares prior to exercise of rights + Total amount received from exercise}}{\text{Number of shares outstanding prior to exercise + Number of shares issued in the exercise}}$$

$$= \frac{(₹\ 25 \times 10,00,000 \text{ shares}) + (₹\ 20 \times 2,50,000 \text{ shares})}{10,00,000 \text{ shares} + 2,50,000 \text{ shares}} = \frac{₹\ 3,00,00,000}{12,50,000 \text{ shares}} = ₹\ 24$$

2. Computation of adjustment factor

$$= \frac{\text{Fair value per share prior to exercise of rights}}{\text{Theoretical ex-rights value per share}}$$

$$= \frac{₹\ 25}{₹\ 24 \text{ (Refer Working Note 1)}} = 1.04 \text{ (approx.)}$$

Calculation of Diluted EPS

Question 8

Net profit for the current year	₹ 1,00,00,000
No. of equity shares outstanding	50,00,000
Basic earnings per share	₹ 2.00
No. of 12% convertible debentures of ₹ 100 each	1,00,000
Each debenture is convertible into 10 equity shares	
Interest expense for the current year	₹ 12,00,000
Tax relating to interest expense (30%)	₹ 3,60,000

Compute Diluted Earnings Per Share.

Answer

Adjusted net profit for the current year (1,00,00,000 + 12,00,000 – 3,60,000) = ₹ 1,08,40,000

No. of equity shares resulting from conversion of debentures: 10,00,000 Shares

No. of equity shares used to compute diluted EPS: (50,00,000 + 10,00,000) = 60,00,000 Shares

Diluted earnings per share: (1,08,40,000/60,00,000) = ₹ 1.81